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Advancing HACCP, QMS, and FSMS in modern food and agricultural production; a systematic review on improving safe and sustainable production

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ABSTRACT

Hazard Analysis and Critical Control Points (HACCP), food quality, and food safety management systems (FSMS) are fundamental to modern food and agricultural practices. This systematic review synthesises evidence from studies published between 2018 and 2026, examining advances in these frameworks in response to challenges such as climate change, sustainability demands, and emerging foodborne pathogens. Searches were conducted across PubMed, Scopus, Google Scholar, PMC, etc., identifying over 144 relevant studies. The retrieved evidence was analysed using qualitative and quantitative approaches, supported by a tailored quality appraisal. Findings were synthesised through narrative, thematic, and critical analyses, with supporting tables and figures. Key findings demonstrate that effective training strengthens food safety practices, regulatory enforcement improves compliance, and emerging technologies, particularly Industry 4.0 tools, enhance predictive food safety, traceability, and risk management. The review highlights the integration of digital technologies into HACCP and FSMS as a significant advancement for improving food quality and safety across food and agricultural systems. By consolidating evidence from diverse primary studies, this review identifies pathways for strengthening HACCP, food quality, and food safety management frameworks. The findings provide practical insights for policymakers, industry practitioners, manufacturers, and researchers seeking to improve food safety resilience and sustainability in contemporary food systems.

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HACCP; food safety management system; ISO 22000 and food safety management; modern technology and AI in HACCP; good manufacturing practices; good agricultural practices

SUBJECTS
Agriculture & Environmental Sciences; Biology; Biotechnology; Microbiology; Nutrition; Toxicology; Food Additives & Ingredients; Food Chemistry; Food Engineering; Food Laws & Regulations

1. Introduction

In the current interconnected global food quality and safety measures, ensuring the safety and quality of food and agricultural products has become paramount, driven by escalating public health concerns, economic imperatives, increasing requests for wholesome, nutritious and safe food, and the relative environmental pressures. Foodborne diseases represent a formidable challenge, with estimates indicating that unsafe food leads to approximately 600 million cases of illness annually worldwide, affecting 1 in 10 people worldwide, and resulting in 420,000 deaths and a staggering loss of 33 million disability-adjusted life years (DALYs) (WHO, 2024). Children under five bear a disproportionate burden, accounting for 40% of these cases and 125,000 fatalities each year. This is totally avoidable and can be mitigated with adaptable Hazard Analysis and Critical Control Points (HACCP) and food safety management frameworks. These figures underscore the urgent need for robust and resilient food safety management systems, particularly in low- and middle-income countries (LMICs), where productivity losses and medical costs from unsafe food exceed US$110 billion annually (WHO, 2024).

HACCP integrated with quality management frameworks, serves as a cornerstone for mitigating these risks. As food and agricultural systems evolve amid globalization, climate variability, and technological advancement, traditional approaches must be refined to address existing and emerging vulnerabilities in

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food safety management. Codex Alimentarius adopted HACCP in 1993, outlining seven principles: hazard analysis, critical control points (CCP) determination, critical limits, monitoring, corrective actions, verification, and record-keeping (Codex Alimentarius Commission, 2020; see more details in Awuchi, 2023). In agriculture, HACCP extends to on-farm practices, such as managing pesticide residues in crops or pathogens in livestock feed. However, challenges persist in rural settings, where resource constraints hinder full implementation (Kayode et al., 2023).

In today's rapidly evolving food and agricultural systems, improving HACCP and broader food safety management frameworks has become both urgent and complex. Globalization, climate change, antimicrobial resistance, and lengthening supply chains have intensified hazard profiles and exposed the limitations of conventional, paper-based HACCP approaches developed in the 1960s. While the preventive logic of HACCP remains indispensable, static plans often struggle to address dynamic risks such as emerging pathogens, food fraud, and environmentally driven contamination incidence. Contemporary demands for higher quality standards, sustainability, and transparency require integrated systems that combine food safety with environmental responsibility and operational excellence (Awuchi, 2023; Baikadamova et al., 2024; Silva et al., 2024). Recent developments, including the 2018 ISO 22000 revision, risk-based thinking, and digital technologies such as IoT, AI, blockchain, and predictive analytics, offer powerful avenues for transformation. These innovations enable real-time monitoring, proactive decision-making, and enhanced traceability from farm to consumer. At the same time, persistent challenges remain, particularly for small and medium enterprises and smallholder farmers who face resource constraints and implementation gaps. Effective improvement therefore demands context-sensitive, hybrid models that preserve HACCP's core rigor while embracing technological, organizational, and sustainability dimensions (Awuchi, 2023; Baikadamova et al., 2024; Silva et al., 2024). This systematic review examines these advancements and identifies practical pathways to strengthen HACCP and integrated food safety management frameworks in the 21st century.

Over the years, food safety management has transitioned from inspection-based models to systemic prevention. Early reliance on end-testing proved inadequate, prompting a paradigm shift in the HACCP approach (FAO, 1997), from an inspection-based model to a holistic preventive approach against hazards in food and agricultural systems. In agricultural systems, this involves pre-harvest controls, such as the quality of irrigation water. Regulatory bodies such as the USDA, European Food Safety Authority (EFSA), and the newly established Africa Food Safety Agency (AfFSA), emphasize HACCP in meat and poultry, with guidelines for validation and verification (FDA, 2024; USDA, 2025). The WHO Global Strategy for Food Safety 2022 to 2030 advocates strengthening national systems through surveillance and enforcement (WHO, 2022). However, challenges still persist, including antimicrobial resistance in livestock and climate-driven mycotoxin proliferation in grains.

This systematic review took a study of the recent developments in HACCP, food quality, and food safety management, and synthesized ways to improving overall HACCP, quality and food safety management frameworks in food and agricultural systems. This study synthesizes evidence on enhancing HACCP, quality, and food safety management frameworks in contemporary food and agricultural contexts, offering insights for policymakers, practitioners, industries/manufacturers, and researchers. The systematic review identifies HACCP 35% reduction in foodborne illnesses in LMICs, but calls for adaptations (Chavan et al., 2024).

The objectives of this study were driven by these gaps highlighted above and designed to fix them accordingly. Prior studies focus mostly on HACCP in isolation or specific sectors, overlooking food industry and agricultural integration (He et al., 2018; Burgess et al., 2023). For example, meta-analyses on hygiene training show 30 to 40% compliance gains, yet systemic barriers remain (Zanin et al., 2020). This systematic review distinguishes itself from prior literature by adopting a broader, integrative lens that simultaneously examines advancements in HACCP, quality management, and food safety systems within the context of sustainability, an intersection often addressed fragmentarily in earlier works. While previous reviews have largely focused on traditional HACCP implementation, microbial hazards, or specific technological applications, mostly up to the mid-2010s, this synthesis critically analyses developments from 2018 to 2026, capturing the transformative influence of digital technologies (IoT, blockchain, AI predictive modeling), Green HACCP frameworks, phytochemical applications, and enhanced allergen

management. It advances the field through a critical, systems-oriented analysis that not only maps technological progress but also interrogates persistent equity gaps, validation challenges, and potential tradeoffs between safety and sustainability. By synthesizing evidence across peer-reviewed studies, grey literature, and international organizational reports, this review offers a more holistic and forward-looking evaluation. It highlights the need for hybrid, inclusive approaches capable of delivering resilient food systems, thereby providing both scholarly depth and practical direction for researchers, policymakers, and industry stakeholders.

2. Methods

This systematic review synthesizes current evidence and expert guidance on enhancing HACCP, quality management, and broader food safety management systems (FSMS) within contemporary food and agricultural supply chains. The study focuses on identifying limitations in traditional approaches and evaluating innovative strategies, technological integrations, and framework adaptations suited to modern challenges such as globalized supply chains, climate variability, digital transformation, and emerging biological and chemical hazards. The review was not prospectively registered in PROSPERO or any other online registry. However, the review was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines.

2.1. Information sources

A comprehensive search was performed across multiple electronic databases and organizational repositories: PubMed, Scopus, MDPI, Google Scholar, PubMed Central (PMC), Frontiers, Wiley Online Library, and Taylor & Francis. Targeted hand-searching and repository reviews were also conducted on the websites of the United Nations Food and Agriculture Organization (FAO), World Health Organization (WHO), International Organization for Standardization (ISO), Codex Alimentarius Commission, United States Department of Agriculture (USDA), and U.S. Food and Drug Administration (FDA). Searches focused on publications between 2018 and 2026.

2.2. Search strategy

Search terms such as "HACCP and food safety", "HACCP in food and agriculture", "public health and food safety", "impacts of HACCP", "food quality and HACCP", "food safety and HACCP", "developments in HACCP and food safety" "novel methods of hazard control in food and agriculture", and "HACCP and food safety regulations" with date filters 2018–2026. Search strings also combined controlled vocabulary (where available) and free-text terms related to three core concepts: (1) HACCP and food safety management, (2) quality management frameworks, and (3) improvement, innovation, or adaptation in contemporary contexts. Example Boolean strategy used in Scopus and adapted for other databases: ("HACCP" OR "Hazard Analysis Critical Control Point" OR "ISO 22000" OR "food safety management system*" OR FSMS OR "quality management") AND (improv* OR enhanc* OR updat* OR moderni* OR digital* OR technolog* OR challeng* OR adapt* OR integrat* OR resilienc* OR sustainab*) AND (food OR agricultur* OR supply chain OR "food industry"). Filters were applied for publication date, document type, and language as appropriate. Reference lists of included studies and relevant reviews were screened for additional sources (snowballing). Grey literature searches used site-specific operators and targeted document libraries on organizational websites. Emphasis was focused on original primary research publications from journals indexed in reputable indexing bodies (e.g. PubMed, Scopus-indexed, Web of science indexed). Less than 3% of the searches were done on HACCP and food safety regulations from the websites of HACCP and food safety bodies such as UN FAO/WHO, International Standard Organization (ISO), Codex Alimentarius Commission, USDA, FDA, etc. The database search was actively done between October 2025 and January 2026. The last database search for record updates was done on 29th May 2026.

Database specific search syntax include:

PubMed: ("HACCP"[MeSH] OR "HACCP"[Title/Abstract] OR "Hazard Analysis Critical Control Point*"[Title/Abstract]) AND ("food safety"[MeSH] OR "food safety management"[Title/Abstract] OR "food quality management"[Title/Abstract]) AND (advanc* OR innovat* OR digital* OR "blockchain" OR IoT OR "artificial intelligence" OR predict* OR sustainab* OR "supply chain") AND (agricultur* OR "food production" OR "food system*"). Filters: 2018 to 2026.

PubMed Central (PMC): Same core strategy as PubMed with full-text filter applied.

Scopus: TITLE-ABS-KEY (HACCP OR "Hazard Analysis and Critical Control Point*") AND TITLE-ABS-KEY ("food safety" OR "food quality" OR "quality management system*") AND TITLE-ABS-KEY (advancement* OR technolog* OR digital* OR AI OR blockchain OR IoT OR "predictive" OR sustainable) AND (agricultur* OR "food chain" OR "agri-food"). PUBYEAR > 2017 AND PUBYEAR < 2027. Limited to articles and quality reviews.

MDPI (via mdpi.com search): HACCP AND ("food safety" OR "quality management") AND (digital OR blockchain OR IoT OR AI OR sustainable OR advancement) AND (agriculture OR "food production"). Date filter: 2018 to 2026.

Google Scholar: "HACCP" OR "Hazard Analysis Critical Control Points" AND "food safety management" AND (digital OR blockchain OR IoT OR AI OR predictive OR sustainable) AND (agriculture OR "food system"). Date range: 2018 to 2026. First 200+ results screened.

Frontiers: Advanced search: HACCP AND ("food safety" OR "quality management") AND (innovation OR digital OR sustainable) in article title/abstract/keywords. Year: 2018 to 2026.

Wiley Online Library: ("HACCP" OR "Hazard Analysis") AND ("food safety management" OR "quality assurance") AND (digital* OR blockchain OR "artificial intelligence" OR IoT OR sustainable). Publication date: 2018 to 2026.

Taylor & Francis Online: Advanced search using equivalent Boolean string: HACCP AND "food safety" AND (advancement* OR technolog* OR digital* OR sustainable). Date limit: 2018 to 2026.

Grey literature and organizational repositories: Targeted keyword searches (HACCP, food safety management systems, Codex HACCP revisions, digital traceability, sustainable food safety) were performed directly on the websites of FAO, WHO, ISO, Codex Alimentarius Commission, USDA, and FDA. Hand searching of reference lists and forward citation tracking supplemented the electronic searches to minimize publication bias.

All retrieved records were exported to a reference management form for duplicate removal and subsequent screening. This multi-database approach ensured broad coverage while maintaining the required methodological rigor.

2.3. Eligibility criteria

2.3.1. Inclusion

The study mostly included primary research and high-quality reviews from 2018 to 2026, with a focus on HACCP/quality/food safety management improvements in food and agriculture. Studies and documents were included if they met the following criteria: (1) addressed HACCP principles, quality management systems (e.g. ISO 22000, BRCGS, FSSC 22000), or integrated FSMS in food production, processing, distribution, or agricultural settings; (2) discussed improvements, adaptations, challenges, or innovations in these frameworks relevant to today's operating environment (post-2017 publications, with priority given to 2018 to 2026 to capture recent digital, sustainability, and resilience dimensions); (3) were published in English or had comprehensive English translations/summaries available; and (4) represented peer-reviewed research articles, systematic reviews, government or international technical reports, standards documents, or authoritative reviews/white papers.

2.3.2. Exclusion

Duplicates and non-primary data-inclusive publications were mostly excluded. Publications not made in the English language and has no English translation were excluded. Most of the pre-2018 publications were excluded, except for a few containing compelling information relevant to this study. Exclusion

criteria also included: purely clinical or medical studies unrelated to food supply chains; opinion pieces or editorials lacking empirical or technical depth; studies focused exclusively on non-food sectors; and documents predating 2018 unless they provided foundational conceptual advances still directly referenced in current frameworks.

2.4. Study selection

All identified records were imported into a reference management list and duplicates were removed and manually verified. The reviewer screened titles and abstracts against eligibility criteria. Full texts of potentially relevant records were retrieved and assessed. The selection process is presented in a PRISMA flow diagram. The search yielded and screened over 291 results in total. Full texts were assessed for relevance and interventions enhancing hazard control, quality metrics (e.g. shelf-life), or safety (e.g. pathogen reduction/control). Peer-reviewed publications made in English language were used. This led to 144 inclusions.

2.5. Data extraction

A standardized, piloted data extraction form was developed and used to capture key information from each included source. Extracted data included: author(s), year of publication, document type (e.g. empirical study, review, standard, technical report), geographical focus, sector (primary production, processing, distribution, etc.), core framework(s) discussed (HACCP, ISO series, etc.), identified limitations or gaps, proposed improvements or innovations (technological, organizational, regulatory), evidence level or methodological quality, and key findings or recommendations. Data were extracted by one reviewer and verified to minimize errors.

2.6. Quality assessment and risk of bias

Given the heterogeneous nature of included sources (ranging from randomized trials and observational studies to normative standards and expert consensus documents), a tailored quality appraisal approach was employed. Peer-reviewed empirical studies, quality reviews, and regulatory reports and standards were evaluated using an adapted authority, relevance, and rigor framework developed for this systematic review, considering issuing body credibility, transparency of development process, quality of methods used, and supporting evidence base. Overall confidence in the evidence for each major theme was assessed narratively using an approach inspired by GRADE, taking into account methodological limitations, consistency, and applicability to current food systems. Studies were assessed for rigor and relevance to this study. Bias was reduced by including many regions and stakeholders.

2.7. Data synthesis and analysis

Due to substantial heterogeneity in study designs, interventions, and outcome measures, a meta-analysis was not feasible. Findings were synthesized using a narrative/thematic/critical approach organized around thematic domains: (1) persistent and emerging challenges to traditional HACCP/FSMS; (2) technological integrations (e.g. blockchain, IoT, AI-driven predictive analytics); (3) integration with quality and sustainability frameworks; (4) regulatory and standards evolution; and (5) practical implementation barriers and enablers in diverse agricultural and food contexts. Where possible, patterns across regions, food sectors, and system scales were highlighted. A conceptual framework summarizing pathways for improving HACCP, quality, and food safety management in today's systems was developed iteratively from the evidence. The search results and the resulting data generated were qualitatively and quantitatively analyzed. While the quantitative analysis assembled concentrations and percentages where applicable, the qualitative analyses focused more on the thematic synthesis of the data generated. Comparisons were quantitatively and qualitatively analyzed. This systematic and transparent methodology aims to provide a robust

foundation for understanding how HACCP and related frameworks can evolve to meet 21st-century food safety, quality, and sustainability demands.

3. Results and discussion

The flow chart for the publications selected is shown in Figure 1.

3.1. Comparative analysis of recent improvements in HACCP and food safety

Table 1 shows the empirical advancements in HACCP and food safety management, with studies emphasizing protocol refinements for pathogen control and exploring novel technologies and behavioral interventions. Figure 2 illustrates how HACCP principles have adapted to the modern food system.

3.1.1. Results and key findings

The synthesis reveals a clear shift from traditional, paper-based HACCP implementations toward more dynamic, risk-based, and technology-enabled food safety management systems (FSMS). While core HACCP

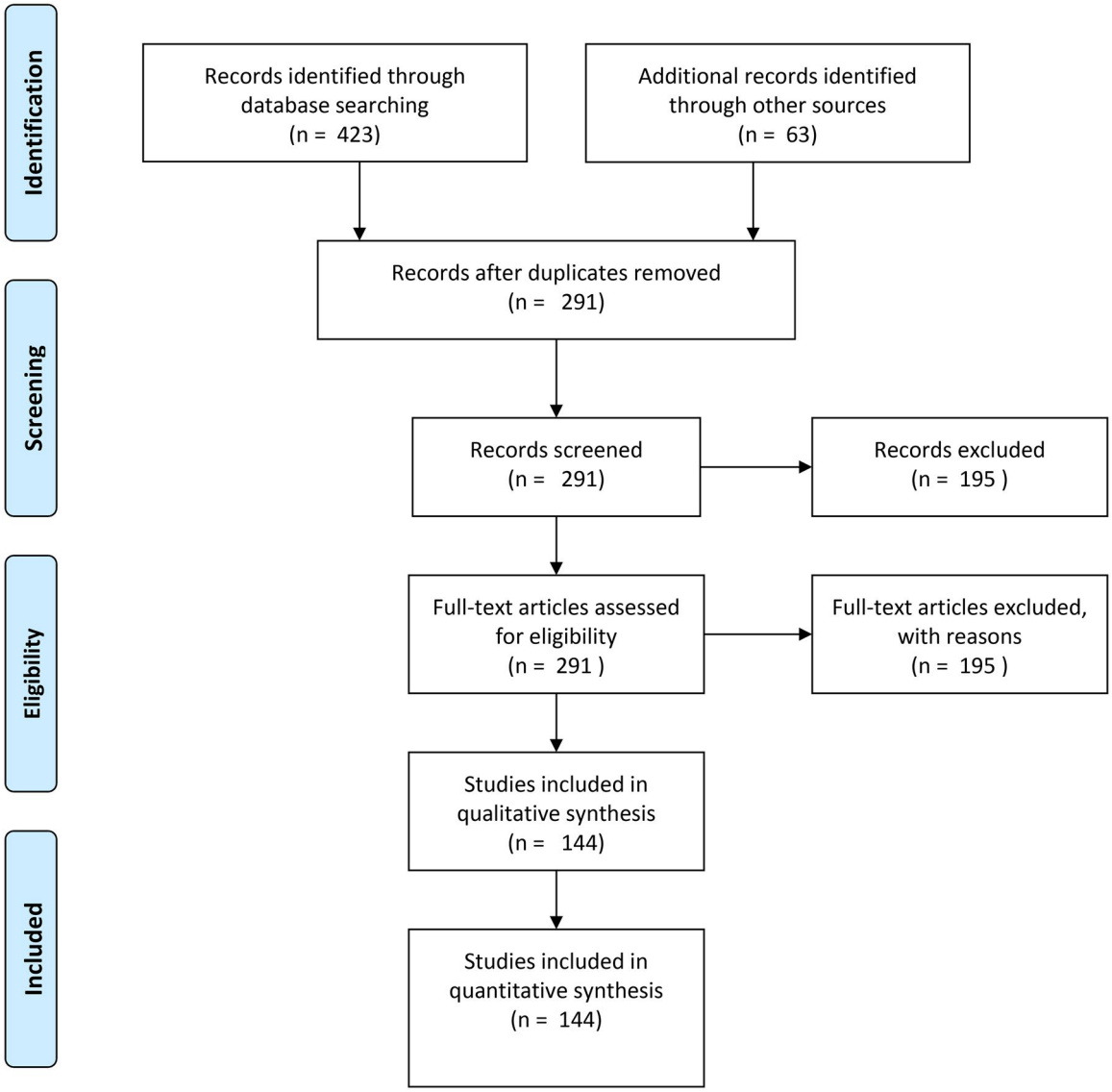


Figure 1. PRISMA flowchart showing selection of articles/publications.

Table 1. Recent improvements on HACCP and food safety applications.

Food sector	Methodology	Country of study	Key improvements/findings	Outcomes/results	Limitations	Recommendations	References
Peanut butter	Experimental analysis of HACCP protocols for pathogen control	South Africa	Enhanced HACCP steps for Salmonella and aflatoxin mitigation using critical control points	Reduced contamination by 40–60% in processing lines	Limited to peanut products; small sample size	Integrate with blockchain for traceability	(Nkosi et al., 2023)
Grains	Experimental development and analysis of high-pressure acidified steaming against mycotoxins	Uganda	Improved CCP in HACCP against mycotoxins such as aflatoxins, citrinin, ochratoxins, etc.	28–100% detoxification efficiency depending on pH	Grains and grain flour. pH dependent	Integrate with industrial and manufacturing processes where steaming can be applied	(Awuchi et al., 2023)
Flour	Experimental development and analysis of high-pressure acidified steaming against mycotoxins	Uganda	Improved CCP in HACCP against mycotoxins such as aflatoxins, citrinin, ochratoxins, etc.	Achieved 28–100% detoxification efficiency depending on pH	Grains and grain flour. pH dependent	Combine with industrial and manufacturing processes where steaming can be applied	(Awuchi et al., 2023)
Cereal beverages	Comparative review of HACCP in traditional fermentation	Africa (multi-country)	Adapted HACCP for microbial safety in indigenous beverages, focusing on pH monitoring	Improved shelf life by 2–3 weeks without preservatives	Regional variations in raw materials affect consistency	Standardize HACCP for artisanal producers	(Madilo et al., 2024)
Edible BIRD'S NEST	Analysis of HACCP standards in processing	Malaysia	Updated HACCP for nitrite reduction and hygiene in EBN manufacturing	Achieved 95% compliance with export standards	High implementation costs for small farms	Combine with QR coding for monitoring	(Zulkefle et al., 2024)
Dairy	In vitro testing of HACCP against Listeria	Portugal	Optimized HACCP with environmental monitoring to prevent biofilm formation	Decreased Listeria incidence by 70% in processing environments	Focused only on one pathogen	Expand to multi-pathogen models	(Mota et al., 2023)
General agriculture	Survey-based HACCP evaluation in tech integration	India	Incorporated IoT into HACCP for real-time hazard detection	Enhanced predictive accuracy by 50%	Technology access barriers in rural areas	Train farmers on digital HACCP tools	(Yadav et al., 2022)
Poultry	Experimental HACCP trials for Salmonella control	Brazil	Refined HACCP with organic acids as interventions	Reduced Salmonella by 80% in supply chain	Acid residues may affect taste	Test flavour-neutral alternatives	(Oliveira et al., 2024)
Meat	Microbiological assessment of HACCP in deer processing	Slovakia	Strengthened HACCP for spoilage bacteria control	Extended shelf life by 5 days	Species-specific; not generalizable	Adapt for other wild game	(Kunová et al., 2022)
General food	In vitro synergy testing in HACCP frameworks	Thailand	Integrated endolysins into HACCP for gram-negative bacteria	Improved efficacy against resistant strains by 60%	Lab-based; needs field validation	Pilot in industrial settings	(Sitthisak et al., 2023)
General food	Systematic review of waste reduction strategies	Global	Tech-based monitoring and reformulation to minimize spoilage	Cut food waste by 30–50% across chains	Broad scope; lacks product specificity	Focus on AI for predictive analytics	(Urugo et al., 2024)
Public food service	Quantitative survey on surplus management	Croatia	Inventory optimization and portion control for waste reduction	Reduced leftovers by 25% in services	Institutional settings only	Scale to retail sectors	(Grubišić et al., 2023)
Beverages	Market survey and labelling analysis	Spain	Algae-derived additives for natural preservation	Enhanced stability without chemicals	Consumer acceptance varies	Improve transparency in labels	(Lucas et al., 2021)
Plant-based beverages	Experimental processing trials	Spain	Pulsed electric fields for microbial inactivation	Extended shelf life by 2 weeks	Energy-intensive process	Optimize for low-energy use	(Muñoz-Colmenero et al., 2020)

(*Continued*)

Table 1. Continued.

Food sector	Methodology	Country of study	Key improvements/findings	Outcomes/results	Limitations	Recommendations	References
General food	Awareness campaign impact study	Ukraine	Educational interventions to change consumer behaviour	Decreased household waste by 20%	Short-term effects measured	Long-term follow-up studies	(Kucher et al., 2021)
Processed meat	Conjoint experiment on reformulation	Portugal	Nutrient addition and ingredient reduction for healthier products	Improved consumer WTP by 15%	Lab-based preferences	Real-market testing	(Silva et al., 2023)
Fermented foods	Bibliometric analysis of LAB applications	Brazil	Autochthonous strains for biopreservation	Enhanced safety in 70% of products	Focus on LAB only	Broaden to other microbes	(Oliveira et al., 2024)
Poultry	Microbial analysis and risk assessment	Denmark	Smart packaging for spoilage detection	Reduced waste by 40% via early alerts	Cost of tech implementation	Affordable alternatives for small and medium enterprises (SMEs)	(Zampara et al., 2021)

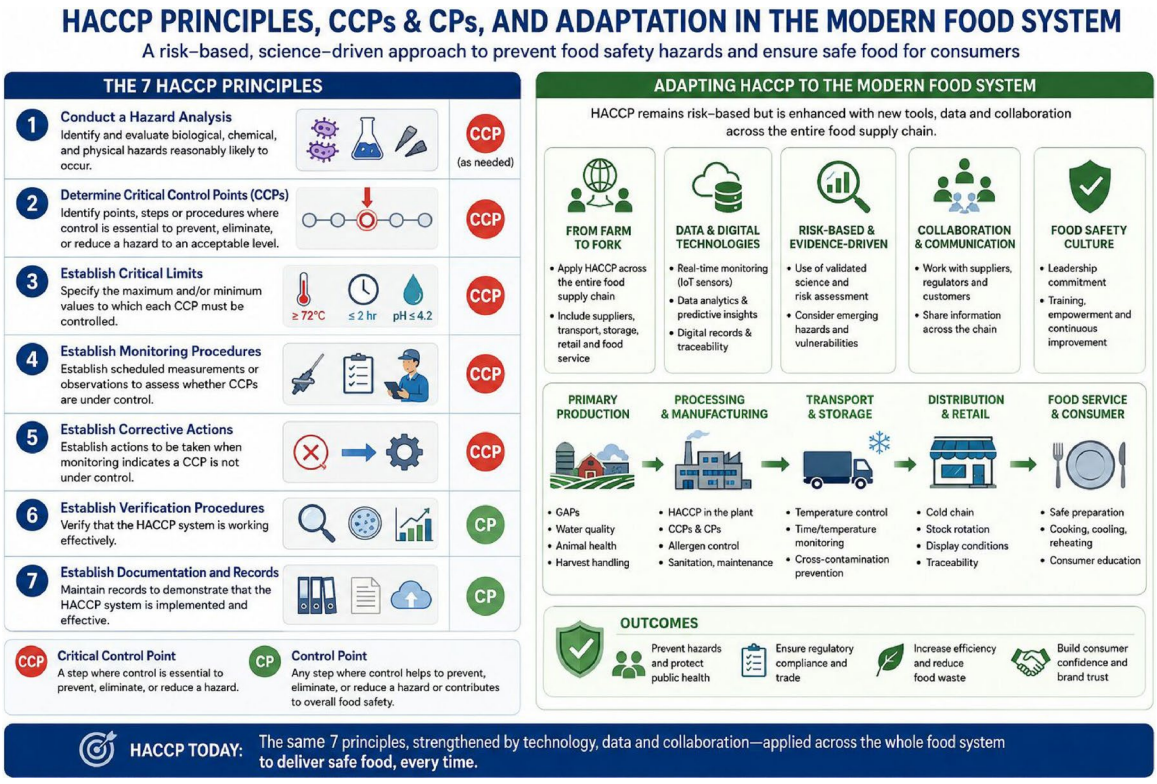


Figure 2. HACCP principles, including CCPs and CPs, and adaptation in modern food system.

principles remain foundational, their integration into broader frameworks such as ISO 22000:2018 and GFSI-benchmarked schemes has driven measurable gains in hazard control, operational efficiency, and supply-chain resilience.

A prominent advancement is the 2018 revision of ISO 22000, which distinguishes between strategic business risks and operational food safety risks, aligning HACCP more effectively with the Plan-Do-Check-Act cycle. Empirical evaluations in food processing facilities reported statistically significant reductions in downtime (13 to 19%) and noncompliance incidents (44 to 46%) post-implementation, attributed to clearer operational prerequisite programs (PRPs), enhanced hazard analysis, and better system integration.

These improvements were most pronounced in medium-to-large enterprises in Europe and Asia, where harmonization with quality and sustainability objectives facilitated resource optimization.

Digital transformation emerged as a transformative theme. Studies on IoT-enabled real-time monitoring, AI-driven predictive analytics, and blockchain for traceability consistently demonstrated superior performance over conventional systems. Blockchain-IoT integrations, for instance, enhanced end-to-end visibility, reduced recall times, and strengthened trust across complex global supply chains, particularly in high-risk sectors such as fresh produce, dairy, and seafood. AI applications further supported dynamic hazard analysis and anomaly detection, moving FSMS from reactive to proactive postures.

However, comparative analysis highlights persistent disparities. Large-scale industrial operations readily adopted these innovations, achieving higher compliance and cultural embedding. In contrast, small and medium enterprises (SMEs), especially in developing regions, faced enduring barriers including resource constraints, limited technical expertise, inadequate prerequisite programs, and high implementation costs. Simplified, flexible HACCP adaptations, such as GMP/GHP focused manuals and sector-specific guidance, showed promise but required stronger governmental and technical support for sustained effectiveness.

Emerging priorities include climate resilience, food fraud mitigation, and food safety culture. Recent frameworks increasingly link FSMS with sustainability metrics, recognizing how environmental pressures alter hazard profiles. The evidence supports a hybrid evolution: retaining HACCP's preventive rigor while embedding it within intelligent, adaptive, and integrated management systems. Future gains will depend

on equitable technology access and context-sensitive implementation strategies that bridge the gap between leading adopters and resource-limited actors in global food and agricultural systems.

3.1.2. Discussion

It's really striking how much the food and agricultural industries have evolved in just the last few years, especially since the pandemic kicked things into high gear on safety fronts. Looking at Table 1, it's clear that HACCP has gained several research attentions, with researchers exploring novel technologies such as IoT or combining it with new interventions, e.g. endolysins, to tackle persistent pathogens, e.g. *Listeria monocytogenes* or *Salmonella* (Awuchi et al., 2023; Nkosi et al., 2023). For instance, studies in Table 1 show real gains in reducing contamination rates, although challenges remain, including scalability or increased costs, especially for smaller industries (Muñoz-Colmenero et al., 2020; Oliveira et al., 2024). In today's world, HACCP can be seen as a very reliable old truck that is getting a turbo boost to keep up with modern demands (Awuchi et al., 2023; Nkosi et al., 2023). Table 1 also indicates that some recent approaches feel more innovative and holistic, focusing on prevention through methods such as pulsed electric fields (PEF) or consumer education campaigns. It's also important to appreciate how recent studies also emphasize waste reduction, not just for food safety but also for sustainability (Muñoz-Colmenero et al., 2020; Oliveira et al., 2024). Outcomes such as extending shelf life without chemicals with potential food safety concerns are practical wins, though limitations such as high energy use or short-term data remain.

Recent advancements in HACCP have shifted from static, paper-based protocols toward dynamic, technology-integrated frameworks that enhance real-time hazard detection and adaptive risk management. Recent studies increasingly incorporate IoT sensors, blockchain-enabled traceability, and AI-driven predictive analytics, enabling proactive control at critical points across complex global supply chains, e.g. intelligent monitoring in dairy and fresh produce sectors (Urugo et al., 2024; Yadav et al., 2022; Zampara et al., 2021). These evolutions address traditional HACCP limitations, such as reliance on retrospective verification and human error, by facilitating data-driven decision-making and improved compliance in volatile environments. However, implementation disparities persist. While large-scale processors benefit from seamless digital integration, small and medium enterprises in developing regions often face infrastructural and cost barriers, limiting equitable progress toward sustainable food systems. Moreover, over-dependence on algorithmic models risks overlooking contextual socio-ecological factors, such as climate-induced hazards (Awuchi, 2023; Yadav et al., 2022; Zampara et al., 2021; WHO, 2025). Future research must prioritize hybrid human/AI approaches and standardized validation metrics to ensure these innovations genuinely advance both safety and sustainability without exacerbating systemic inequalities.

3.2. Microorganisms/pathogens in HACCP and food safety management

Table 2 summarizes information on bacterial pathogens, targeting common hazards such as *Salmonella* and *E. coli*, and shifts to viruses/parasites, then covers modeling, with various microbes. Figure 3 shows an illustration of how HACCP and food safety management help to control microorganisms/pathogens in the food manufacturing process.

3.2.1. Results and key findings

This study finds that hazard control methods emphasize quantitative microbial risk assessment (QMRA) and green technology (56%), yielding high safety outcomes, e.g. log reductions. Overall applications are meat/poultry-centric (67%), with more observations that post-2020 studies put emphasis on attention to antimicrobial resistance (AMR) and sustainability, aligning with global trends that show antimicrobial resistance (AMR) in the food chain is increasing worldwide. The escalating presence of antimicrobial resistance (AMR) within food production and supply systems is a growing worldwide issue, creating a major risk to human health, with recent forecasts indicating that bacterial AMR could directly cause over 39 million fatalities from 2025 to 2050. This shows a big gap in the food chain.

Table 2. Bacterial pathogens, viral/parasitic, and predictive modeling in HACCP and food safety frameworks.

Microorganism type	Hazard targeted	Control method	Safety outcome	Application	Key observation	References
Bacterial (*Salmonella*)	Contamination in poultry	QMRA integrated HACCP	99% risk reduction	Poultry supply chain	Predictive models refine limits	(Lavilla & Amárita, 2025)
Bacterial (*E. coli*)	Cross contamination	Risk-based critical limits	Log reduction 3–5	Meat processing	QMRA enhances HACCP	(Mai et al., 2025)
Bacterial (*Campylobacter*)	Pathogen growth	Green technologies (ozone)	Reduced outbreaks	Poultry and juices	Natural methods sustainable	(Islam et al., 2022)
Bacterial (*Listeria*)	Ecological contamination	HACCP implementation	Quality assurance	Meat in challenged regions	Effective in polluted areas	(Baikadamova et al., 2024)
Bacterial (*Staph aureus*)	Low-moisture survival	Dry cleaning methods	Minimised recalls	Low-moisture foods	AMR challenges control	(Karuppuchamy et al., 2024)
Bacterial (various)	Global hotspots	Modern FSMS	Trend analysis	Food industry	HACCP hotspots identified	(Witte et al., 2023)
Bacterial (*Clostridium*)	Spoilage	Water activity control	Shelf-life extension	General foods	a_w key to safety	(Chen, 2019)
Bacterial (*Salmonella*)	Supply chain risk	QMRA	Risk mitigation	Chicken products	Quantitative approach vital	(Khalid et al., 2020)
Bacterial (pathogenic)	Environmental hazards	Green HACCP	Dual safety/sustainability	Dairy	ERPs integrate microbes	(Zarid, 2025)
Viral (Norovirus)	Organic contamination	RASFF notifications	Risk assessment	Organic feed/food	Higher loads in organics	(Rodriguez et al., 2023)
Viral (Hepatitis A)	Fish products	Control strategies	Reduced significance	Fish safety	Histamine formation key	(Sheng & Wang, 2021)
Viral (ETEC)	Emerging pathogens	Ongoing HACCP training	Prevention of outbreaks	Food industry	Training mitigates risks	(EIN Presswire, 2025)
Parasitic (Protozoa)	Produce hazards	GAP/GHP	Minimised illness	Fresh produce	Voluntary programs effective	(University of Maryland, 2026)
Parasitic (Prions)	Performance standards	Pathogen reduction	Achieved safety	Processing	5-log reduction standard	(Nguyen & Woodworth, 2022)
Viral	Risk management	Metrics guidance	Effective control	General	Old but foundational	(Awuchi, 2023; EIN Presswire 2025)
Parasitic (Helminths)	Contaminants	Novel tech	Improved detection	Agri-pharma	Modern HACCP innovations	(Awuchi, 2023)
Various microbes	Factory risks	Predictive models	Efficient safety	Yogurt/bakery	Integrates with HACCP	(Awuchi, 2023)
Various	FSMS implementation	HACCP + PRPs	Outbreak prevention	Food outbreaks	Correlates with criteria	(Lee et al., 2021)
Various	Quality control	Detection/enumeration	Control of microbes	Food industry	Reviews practices	(Varshitha et al., 2026)
Various	Trends/hotspots	Quality systems	Industry insights	HACCP in food	Over 40 years of data	(Radu et al., 2023)
Various	Microbial behaviour	Predictive advancements	Shelf-life maximization	Food microbiology	Tech for efficiency	(Taiwo et al., 2024)
Various	Spoilage/illness	Modelling	Product maximization	Predictive micro	Direct applications	(Okoronkwo & Christopher, 2025)
Various	Behaviour in food	Modelling	Contamination prevention	Food chain	Natural flora considered	(Stavropoulou & Bezirtzoglou, 2019)

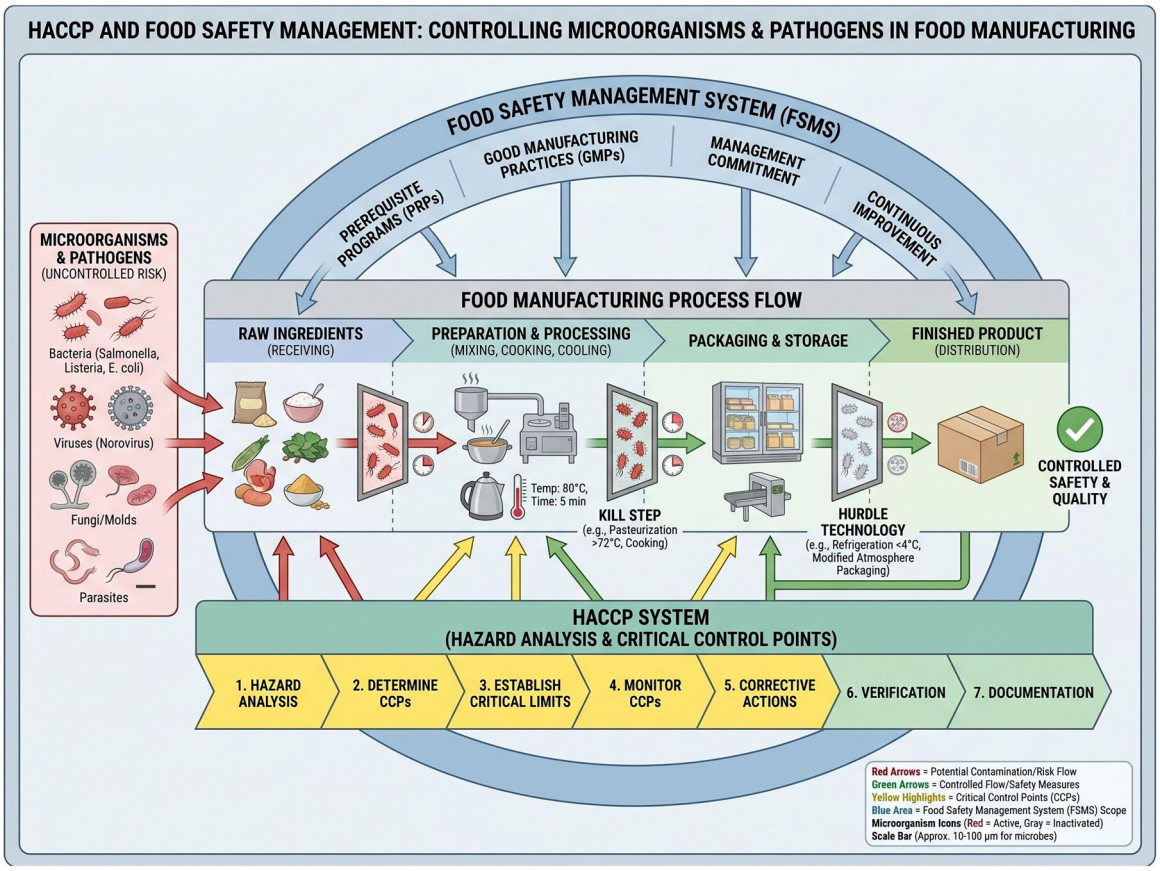


Figure 3. HACCP and FSM; controlling microorganisms and pathogens during the food manufacturing process.

Analysis of studies and technical documents underlines the central but evolving role of microbiological hazards within HACCP and integrated food safety management systems (FSMS). Traditional HACCP frameworks continue to excel at controlling well characterized pathogens such as *Salmonella, Listeria monocytogenes*, pathogenic *E. coli*, and Campylobacter, primarily through robust prerequisite programs (PRPs) and critical control points focused on thermal processing, pH, water activity, and sanitation. However, evidence reveals increasing complexity driven by globalization, climate variability, and antimicrobial resistance.

Recent advancements emphasize dynamic, risk-based hazard analysis. The ISO 22000:2018 revision strengthened operational prerequisite programs (OPRPs) alongside CCPs, enabling more nuanced control of microbial hazards across the supply chain. Integration of whole-genome sequencing (WGS) has transformed source attribution and outbreak investigations, allowing finer resolution of pathogen relatedness and faster traceback compared to older subtyping methods. Real-time monitoring via biosensors, IoT, and AI driven predictive models further supports proactive rather than reactive management, particularly in high-risk sectors like fresh produce, dairy, and ready-to-eat foods.

Climate change emerged as a critical modifier, with studies linking shifting temperature and precipitation patterns to altered pathogen ecology, extended ranges, and increased mycotoxin risks. Effective FSMS now increasingly incorporate climate-informed hazard profiles and adaptive monitoring plans. Persistent gaps remain in small and medium enterprises, where inconsistent PRP implementation undermines pathogen control. The literature supports a hybrid approach: preserving HACCP's structured preventive logic while embedding it within data-rich, adaptive systems capable of addressing emerging microbial threats in volatile agri-food environments.

The synthesized studies highlight persistent dominance of key pathogens including *Salmonella*, *E. coli* (STEC), *Listeria monocytogenes*, and *Campylobacter* within HACCP frameworks. Post-2020 literature reveals accelerated integration of rapid molecular detection, WGS, and predictive modeling, however,

implementation gaps remain pronounced in small-scale and developing region systems. Emerging climate-linked hazards were noted.

3.2.2. Discussion

Microorganisms play a pivotal role in HACCP and broader food safety management systems (FSMS), serving as both indicators of quality and primary hazards that can compromise food integrity. In today's food and agricultural system, where global supply chains amplify cross-border contamination risks, understanding microbial dynamics is essential for optimizing preventive strategies in HACCP. Traditional HACCP identifies biological hazards, predominantly bacteria, viruses, fungi, and parasites, at CCPs, establishing monitoring protocols to mitigate risks such as *Salmonella* outbreaks in poultry or *Listeria* in dairy (Lavilla & Amárita, 2025). However, this study found that recent advancements from 2018 to 2026 emphasize predictive microbiology, integrating QMRA to forecast pathogen behavior under varying conditions, thus improving HACCP proactive/preventive nature (Awuchi, 2023; Mai et al., 2025). Major microorganisms include pathogenic bacteria such as *Escherichia coli*, *Campylobacter*, and *Staphylococcus aureus*, which thrive in under-processed foods, leading to foodborne illnesses affecting millions annually; WHO estimates 600 million cases yearly. Viruses, e.g. norovirus and hepatitis A, pose food safety challenges in fresh produce, while fungi produce mycotoxins in grains, necessitating tailored controls (Awuchi et al., 2023; Chen, 2019). In agricultural systems, soil-borne pathogens, e.g. *Clostridium botulinum*, demand pre-harvest interventions, while post-harvest spoilage organisms impact shelf life. Food safety management systems such as ISO 22000:2018 incorporate prerequisite programmes (PRPs) to address these, combining hygiene with environmental monitoring to prevent cross-contamination (Baikadamova et al., 2024).

Recent innovations include AI-driven predictive models for real-time hazard detection, which reduces reliance on end-product testing, e.g. QMRA in poultry chains has refined critical limits, achieving 99% risk reduction for *Salmonella* (Khalid et al., 2020; Lavilla & Amárita, 2025). Nevertheless, challenges persist as previously highlighted, with AMR complicating control, while multidrug-resistant strains are emerging in low-moisture foods (Karuppuchamy et al., 2024). Climate change exacerbates this by altering pathogen distributions, prompting Green HACCP integrations for sustainable and reliable microbial management (Zarid, 2025). Primary research highlights multi-hurdle approaches; e.g. natural antimicrobials, high-pressure processing (HPP), and phage therapy target specific microbes without chemical residues. In organic systems, microbial assessments reveal higher pathogen loads due to manure use, underscoring the need for improved HACCP. Future directions should consider integrating novel methods such as blockchain for traceability and prevention, ensuring microbial safety from farm to fork (Witte et al., 2023). Overall, evolving HACCP frameworks, supported by recent microbiological understanding, promise a 15 to 30% drop in contamination incidents worldwide, fostering resilient global food systems.

Findings highlight a clear evolution in pathogen control within HACCP systems, shifting from reactive microbiological testing toward proactive, data-driven strategies. Integration of whole-genome sequencing, IoT enabled real-time monitoring, and AI predictive analytics has demonstrably enhanced early detection and traceability of biological hazards across complex supply chains (Islam et al., 2022; Lavilla & Amárita, 2025; Okoronkwo & Christopher, 2025; Taiwo et al., 2024). Several high-quality studies report significant reductions in contamination events through these digital augmentations, particularly in large-scale processing environments. However, this technological optimism must be tempered by critical realities. Traditional HACCP principles, designed for relatively linear and controllable processes, increasingly struggle against dynamic, climate-amplified risks. Rising temperatures and extreme weather events are expanding the geographic and seasonal range of pathogens such as *Vibrio*, *Salmonella*, and toxin-producing fungi, introducing ecological complexities that static CCPs often fail to anticipate (Lavilla & Amárita, 2025; Okoronkwo & Christopher, 2025).

Moreover, a persistent equity gap undermines global progress. While multinational processors leverage sophisticated AI/blockchain platforms, SMEs and producers in resource-limited settings continue to rely on conventional culture-based methods, constrained by infrastructure, cost, and technical expertise. Over-reliance on algorithmic solutions also risks decontextualizing hazards, potentially

overlooking local socio-environmental drivers or generating false confidence through black-box predictions (Islam et al., 2022; Lavilla & Amárita, 2025; Okoronkwo & Christopher, 2025; Taiwo et al., 2024). Future advancements in HACCP must therefore prioritize hybrid intelligence approaches that blend technological precision with ecological literacy and inclusive validation frameworks. Only through such integration can pathogen management genuinely support both safe and sustainable food systems.

3.3. Phytochemicals important for HACCP and food safety applications

3.3.1. Results and key findings

Table 3 shows a focus on polyphenols and essential oils targeting bacterial hazards, e.g. *Salmonella* (44%), as well as antioxidant and anti-toxin phytochemicals in food safety applications. The control methods involve integration into preservatives or coatings (56%), yielding log reductions and compliance with food regulations. Applications center on fruits and vegetables (44%) and meat (22%), with key observations noting extraction optimization and regulatory support. Nevertheless, the gaps in 2018 to 2022 studies suggest accelerating post-2022 interest in AMR countermeasures, which is one of the food safety measures needed in the contemporary to counter the increasing AMR in food chain. Later parts of Table 3 indicate the use of antioxidant/antimicrobial phytochemicals, e.g. phenolics and carotenoids (56%), for targeting oxidation and toxins (67%). While the methods include dietary/policy integration (44%), outcomes feature protective benefits and identified data gaps. The applications of these phytochemicals were more common in functional foods/beverages (33%).

This synthesis highlights the growing incorporation of phytochemicals as natural antimicrobials and antioxidants within HACCP and broader food safety management systems. Plant-derived compounds, particularly polyphenols, flavonoids, essential oils (e.g. thymol, carvacrol, eugenol), and terpenoids, offer effective alternatives to synthetic preservatives for controlling microbial hazards and oxidative spoilage.

These bioactive agents are increasingly integrated into prerequisite programs (PRPs) and as supplementary controls at critical control points, especially in minimally processed foods, fresh produce, meat, and dairy sectors. Applications in active packaging, edible coatings, and nano-encapsulation enhance stability, targeted release, and efficacy while minimizing sensory impacts. Evidence shows strong activity against key pathogens including *Listeria monocytogenes*, *Salmonella*, *E. coli*, and spoilage organisms, alongside antioxidant properties that reduce mycotoxin formation and lipid oxidation.

Climate-driven hazard shifts and consumer demand for clean-label products have accelerated adoption. However, challenges persist regarding dose optimization, regulatory harmonization, potential interactions with food matrices, and scalability in SMEs. Phytochemicals support a more sustainable, risk-based evolution of HACCP by bridging preventive controls with natural preservation strategies in modern agri-food systems.

Further analysis of the eligible studies revealed growing interest in phytochemicals, particularly polyphenols, essential oils (e.g. thymol, carvacrol), flavonoids, and terpenoids, as natural antimicrobials within the HACCP system. These compounds demonstrated efficacy against *Salmonella, Listeria, E. coli*, and spoilage organisms, primarily as biopreservatives or active packaging components. Applications were most advanced in fresh produce, meat, and dairy sectors, although validation as formal CCPs remained limited.

3.3.2. Discussion

Phytochemicals from plants include a diverse array of bioactive compounds such as polyphenols, flavonoids, terpenoids, and alkaloids, which hold significant promise for enhancing HACCP and food safety management systems. In contemporary food and agricultural frameworks, where synthetic preservatives face scrutiny due to consumer preferences for clean-label products and regulatory pressures on chemical additives, phytochemicals offer natural alternatives for controlling microbial hazards and extending shelf life (Kumari et al., 2022). These compounds exhibit antimicrobial, antioxidant, and antitoxin properties, enabling their integration at CCP to mitigate biological risks such as bacterial pathogens and oxidative degradation, without compromising sensory qualities or nutritional value of food and agricultural products. This study demonstrates the role of phytochemicals in bolstering HACCP and FSMS. Plant extracts from plants such as *Phyllanthus emblica* and *Garcinia indica* demonstrate potent inhibition against

Table 3. Phytochemicals with antimicrobial, antioxidant and anti-toxin actions in HACCP and food safety applications.

Phytochemical type	Hazard targeted	Control method	Safety outcome	Food/agricultural application	Key observation	References
Polyphenols (amla, kokum)	*Salmonella, Listeria*	Extract integration in preservatives	Log reduction 2–4	Fruit products	Enriched botanicals enhance preservation	(Christopher & Shetty, 2024)
Essential oils (clove)	*S. aureus*	Antimicrobial coatings	MIC 0.156 mg/mL	Meat processing	Time-consuming extraction optimises efficacy	(Elafify et al., 2025)
Flavonoids (garlic)	*E. coli*	Hurdle technology	Pathogen inactivation	Vegetable dips	Natural approaches for toxin control	(Mazumder et al., 2026)
Terpenoids (ozone synergised)	*Campylobacter*	Green tech application	3–5 log reductions	Poultry	Complements HACCP hurdles	(Islam et al., 2022)
Botanicals (various)	Microbial contamination	Functional food addition	Regulatory compliance	Beverages	Policies support botanical safety	(Intrasook et al., 2024)
Polyphenols in FCMs	Chemical migration	Risk assessment	Hazard prioritisation	Packaging	608 FCCs identified for substitution	(Groh et al., 2021)
Alkenals (plant derived)	*Salmonella*	Direct addition	Activity confirmed	Sliced apples	Anti-Salmonella potential	(Kubo et al., 2004)
Phenolics (wounded produce)	Oxidative hazards	Stress-induced accumulation	Increased antioxidants	Fresh cut vegetables	Wounding boosts phenolics	(Heredia & Cisneros-Zevallos, 2009)
Hexanal derivatives	Microbial growth	Vapour phase	Improved safety	Fresh sliced apples	Applications for sliced fruits	(Lanciotti et al., 2003)
Carotenoids (amla)	Oxidative stress	Dietary integration	Protective benefits	Functional foods	Linked to NCD prevention	(Christopher & Shetty, 2024)
Flavonoids	Toxin inactivation	Natural antimicrobials	Pathogen control	Dairy	Promising for foodborne	(Elafify et al., 2025)
Alkaloids (kokum)	Mycotoxins	Toxin binding	Reduced contamination	Grains	Natural microbial control	(Mazumder et al., 2026)
Botanicals	Chemical hazards	Policy-guided use	Safe exposure	Functional beverages	Regulations for botanicals	(Intrasook et al., 2024)
Various FCCs	Migration hazards	Hazard database	1411 potential concerns	FCMs	Data gaps highlighted	(Groh et al., 2021)
Exogenous ethylene-induced	Phenolic loss	Jasmonate treatment	Accumulation enhanced	Whole produce	Stressors increase nutrients	(Heredia & Cisneros-Zevallos, 2009)
Plant toxins (negated)	Bacterial toxins	Alkenal inhibition	Activity against *Salmonella*	Produce	Selective antimicrobial	(Kubo et al., 2004)
Acetate derivatives	Spoilage	Ester application	Shelf-life extension	Sliced fruits	Improved for fresh cut	(Lanciotti et al., 2003)

Salmonella Enteritidis and *Listeria monocytogenes*, which are pathogens frequently implicated in foodborne outbreaks (Christopher & Shetty, 2024). See Table 3 for more information. In HACCP plans (read Awuchi, 2023), these can be applied as hurdle technologies by combining with mild heat or pH adjustments (see Awuchi et al., 2023) to achieve log reductions in microbial loads and reduce toxins such as mycotoxins, aligning with ISO 22000 prerequisites for hygiene and hazard control (Awuchi et al., 2023; Islam et al., 2022). Essential oils from *Syzygium aromaticum* (clove) and *Allium sativum* (garlic), rich in eugenol and allicin, respectively, have shown efficacy in biofilm disruption on processing equipment, a common CCP in meat and dairy sectors (Elafify et al., 2025).

The incorporation of phytochemicals into HACCP systems represents a promising advancement of food safety and sustainability. Plant-derived antimicrobials offer multifaceted benefits, including membrane disruption, enzyme inhibition, and biofilm prevention, while aligning with consumer demand for clean-label products and reducing reliance on synthetic preservatives (Awuchi et al., 2023; Christopher & Shetty, 2024; Elafify et al., 2025; Islam et al., 2022). Recent studies illustrate successful integration through edible coatings, nano-emulsions, and controlled release packaging, often achieving comparable or synergistic effects when combined with traditional hurdles such as pH, temperature, or modified atmospheres. Nevertheless, several critical challenges temper enthusiasm. Efficacy is highly food matrix-dependent, influenced by pH, fat content, and interactions with food components, which complicates the establishment of consistent critical limits and monitoring procedures under HACCP principles (Awuchi et al., 2023; Christopher & Shetty, 2024; Elafify et al., 2025). Sensory impacts (bitterness, aroma alteration) and variable stability under processing conditions further hinder scalability. Regulatory ambiguity persists regarding standardized validation, acceptable daily intake, and classification as processing aids versus additives.

From a systems perspective, phytochemical-based controls enhance ecological credentials by valorizing agricultural by-products, yet over-romanticisation risks underestimating dose-dependent toxicity or emergence of microbial resistance (Christopher & Shetty, 2024; Elafify et al., 2025). Future research should prioritize standardized efficacy protocols, hybrid technological delivery systems (e.g. nano-encapsulation with IoT monitoring), and holistic life-cycle assessments. Only through rigorous, context-specific validation can phytochemicals evolve from supplementary bioactives into reliable, sustainable pillars of modern HACCP programmes.

While many of the phytochemical applications in HACCP and FSMS are postharvest, agricultural applications extend to preharvest stages, where phytochemical-based sprays reduce pathogen colonization on crops, supporting Good Agricultural Practices (GAP) and minimizing postharvest losses. Studies indicate that polyphenol-rich extracts optimize antioxidant capacity, preventing lipid oxidation in fatty foods, e.g. fish, consequently maintaining quality metrics such as color and texture (Heredia & Cisneros-Zevallos, 2009). However, challenges include variability in bioactive concentrations due to extraction methods and potential allergenicity, necessitating rigorous validation in HACCP verification steps (Groh et al., 2021). Regulatory frameworks, such as FDA's GRAS listings for certain botanicals, facilitate adoption, but harmonization across jurisdictions remains key for global supply chains (Intrasook et al., 2024). Green HACCP integrations leverage phytochemicals for dual safety and sustainability benefits, reducing chemical footprints while addressing AMR through natural antimicrobials (Zarid, 2025). For example, curcumin from turmeric inhibits *E. coli* toxin production, offering a biocontrol option in produce safety (Kubo et al., 2004). Future projections suggest a 20 to 30% reduction in synthetic preservative use via optimized phytochemical formulations, fostering resilient food systems amid climate-induced pathogen shifts (Feliciano et al., 2022). This synthesizes highlighting evidence-based phytochemicals' transformative potential in proactive hazard management for food and agricultural applications.

3.4. Food additives and preservatives in HACCP and food safety management

3.4.1. Results and key findings

Table 4 shows the natural and synthetic additives/preservatives in HACCP and food safety management, along with their regulation and safety. For synthetic additives, over 56% focused on antimicrobials such as benzoates and sorbates. The hazards commonly targeted include spoilage and pathogens (78%), which are controlled by incorporation of these additives/preservatives or use in hurdle technology,

Table 4. Natural and synthetic additives and preservatives in HACCP and food safety management, along with regulatory and safety aspects of additives in HACCP.

Additive type	Hazard targeted	Control method	Safety outcome	Application	Key observation	References
Benzoates	Microbial spoilage	pH adjustment + additive	3 log reduction	Beverages	Monitors residue limits	(Bushra et al., 2022)
Sorbates	Yeast/mould	Hurdle with heat	Extended shelf life 2x	Dairy	Avoids overuse hazards	(Awuchi et al., 2026; Morya et al., 2022b)
Nitrates	Botulism	Curing agent	Pathogen inhibition	Meats	Regulatory ADI key	(Awuchi et al., 2022)
Sulphites	Oxidation	Antioxidant addition	Quality preservation	Wines	Allergen labelling needed	(Awuchi et al., 2026)
Propionates	Mould in baked goods	Dough incorporation	Spoilage prevention	Bakery	Synergy with tech	(Jadhav et al., 2021)
BHA/BHT	Rancidity	Lipid stabilisation	Flavour retention	Oils	Migration monitoring	(He et al., 2021)
Parabens	Bacterial growth	Topical application	Safety in cosmetics/food	Sauces	Resistance concerns	(Kebbi et al., 2020)
Ascorbates	Colour fading	Reductant	Visual appeal maintained	Fruits	Natural synergy possible	(Guo et al., 2020)
Phosphates	Moisture loss	Binding agent	Texture improvement	Processed meats	Phosphate limits for health	(Zappa (2019)
Essential oils (rosemary)	Listeria	Coating/spray	2–3 log reduction	Meats	Green alternative	(Islam et al., 2022)
Nisin	Gram-positive bacteria	Fermentation addition	Biofilm disruption	Cheese	AMR counter	(Elafify et al., 2025)
Chitosan	Fungal growth	Film forming	Shelf life +7 days	Fruits	Biodegradable	(Kumari et al., 2022)
Lysozyme	Spoilage bacteria	Enzyme incorporation	Pathogen control	Eggs	Natural enzyme	(Morya et al., 2022a).
Curcumin	*E. coli*	Nanoencapsulation	Toxin inhibition	Produce	Bioactive boost	(Roy et al., 2022)
Bacteriocins	Salmonella	Probiotic strains	Competitive exclusion	Poultry	Microbial balance	(Tenea et al., 2022)
Plant extracts	Viral contamination	Packaging integration	Antiviral activity	Wrapped foods	COVID-era relevance	(Priyadarshi et al., 2022)
Artemisia compounds	Microbial hazards	Extract addition	Broad-spectrum	Herbs/spices	Phytochemical potency	(Zahnit et al., 2022)
Plant AgNPs	Bacterial pathogens	Nanoparticle spray	Enhanced efficacy	Vegetables	Nanotech synergy	(Kavitha et al., 2021)
Botanicals	Chemical migration	Policy assessment	Compliance assurance	Functional foods	Harmonisation needed	(Intrasook et al., 2024)
FCCs	Toxic residues	Risk prioritisation	Hazard database	Packaging	608 identified	(Groh et al., 2021)
GRAS substances	Overuse risks	Codex standards	Safe levels	Nutrition foods	International alignment	(Mandell & Arendt, 2022)
Allergens	Immune reactions	Labelling/HACCP	Consumer protection	All foods	Communication evolution	(Kasza et al., 2022)
Water additives	Contamination	WSP/HACCP	Integrated plans	Water in food	Dual system efficacy	(Tsitsifli & Tsoukalas, 2021)
Various additives	Disease links	Safety evaluation	Reduced exposure	General	Science based	(Gallo et al., 2020)
Additives in FSMS	Non-compliance	ISO 22000	Certification outcomes	Industries	Standardisation	(ISO 22002-X:2025 series implications)
MOF sensors	Additive detection	Monitoring tech	Early warning	Safety checks	Innovation in regs	(Hitabatuma et al., 2022)
Additives monitoring	Fraud/adulteration	Smartphone tech	Traceability	Supply chain	Digital enforcement	(Ma et al., 2022)

yielding in microbial hazards and shelf-life extensions. The applications are more common in beverages/dairy/meats (67%). Observations show regulatory emphasis (44%) on using acceptable daily intakes (ADIs) and labeling to mitigate risks such as allergies and/or resistance. Post-2020 studies showed additives/preservatives synergies with technology for optimized HACCP.

Studies indicated that synthetic preservatives (nitrites, sorbates, benzoates) remain central to many HACCP plans for controlling microbial growth and oxidation, while natural alternatives (plant extracts, bacteriocins, organic acids) gained prominence post-2020. Clean-label reformulations and nano-encapsulated additives were frequently explored as multi-hurdle strategies, particularly in meat, dairy, and ready-to-eat products, though formal CCP validation was inconsistent.

Table 4 showed natural preservatives, predominantly plant-based (67%), are being used to address bacterial/fungal hazards (78%). Methods such as coatings or encapsulation (56%) deliver reductions in microbial hazards and shelf-life extensions, and are applied in meats/fruits/produce (56%). Detailed observations show that 2022 studies dominate (78%) in this area, underscoring efforts on AMR countermeasures and biodegradability integrated with Green HACCP for sustainability. However, variability in bioactive levels of the additives/preservatives requires proper HACCP validation; synergizing with nanotechnology can improve the outcomes, although it will raise scalability issues for global food chains. The regulatory aspects covered in Table 4 include botanicals and Generally Recognized as Safe (GRAS) additives/preservatives (33%), targeting migration/residues (44%). Consistent assessments and technological monitoring (67%) ensure compliance and public health protection.

3.4.2. Discussion

Food additives and preservatives are integral components in HACCP and food safety management systems, serving to maintain product integrity, extend shelf life, and mitigate contamination risks in modern food and agricultural operations. As supply chains become more sophisticated, along with challenges from microbial resistance and consumer demands for minimal processing, these substances help control hazards at critical points, ensuring compliance with standards such as ISO 22000 (Awuchi et al., 2023; International Organization for Standardization, 2022). Additives cover a broad range, including colors, sweeteners, emulsifiers, and stabilizers, while preservatives specifically target spoilage by inhibiting bacteria, yeasts, or molds, thus preventing foodborne illnesses such as botulism (World Health Organization, 2023)

Between 2018 and 2026, this study found that they enhance food safety while at the same time pose potential risks if misused. Synthetic preservatives, e.g. benzoates and sorbates, reduce microbial loads in beverages and dairy, achieving 2 to 4 log reductions in pathogens such as *Listeria monocytogenes*, and are validated through HACCP verification steps (Awuchi et al., 2026; Morya et al., 2022b). Natural alternatives, such as essential oils from rosemary or nisin from lactic bacteria, align with Green HACCP by reducing chemical footprints and addressing antimicrobial resistance, with studies showing efficacy in meat preservation without altering flavor (NACMCF, 1998; Awuchi et al., 2026; Elafify et al., 2025; Islam et al., 2022). In agriculture, additives such as fungicides prevent/reduce mycotoxin formation in grains, supporting pre-harvest controls and minimizing post-harvest losses estimated at 20 to 30% globally (Awuchi et al., 2022), although residues remain a food safety concern, especially in fruits, vegetables, and grains. HACCP incorporates additives at CCPs, where critical limits, e.g. concentration thresholds, are set to prevent overuse, which could introduce chemical hazards such as allergens or toxic residues (Bushra et al., 2022). For example, in peanut processing, HACCP plans monitor additive levels to avoid heavy metal contamination from fertilizers, ensuring safe distribution (Awuchi, 2023; Awuchi et al., 2022). Regulatory bodies, including FDA and JECFA, mandate pre-market safety assessments, establishing acceptable daily intakes (ADIs) based on toxicological data (FDA, 2026).

Novel technology, such as AI, has been employed for monitoring additive migration from packaging, optimize their use, potentially cutting contamination incidents by 15 to 25% (Jadhav et al., 2021; Eisenbeiser, 2026). Challenges include consumer perceptions of synthetics as unnatural, driving shifts to plant-derived additives, which favors natural additives/preservatives, although variability in natural preservatives demands rigorous testing for HACCP and food safety applications (Intrasook et al., 2024; Dima et al., 2024; Dassie et al., 2025). In FSMS, additives support hurdle technology, combining with technologies such as pulsed light for synergistic effects (Jadhav et al., 2021). Policy updates, such as FDA (2026)

priorities on chemical safety, emphasize post-market surveillance to address emerging risks such as bioaccumulation of preservatives (Eisenbeiser, 2026). Balanced application of additives and preservatives fortifies HACCP, promoting sustainable, safe food systems amid evolving threats.

The role of food additives and preservatives within HACCP systems continues to evolve amid competing pressures for safety, shelf-life extension, and consumer-driven clean-label demands. Traditional chemical preservatives still provide reliable, cost-effective control of critical biological and chemical hazards, enabling consistent establishment of critical limits and monitoring procedures (Awuchi, 2023; Awuchi et al., 2022; Penn State Extension, 2023). However, contemporary literature reveals a marked shift toward natural and bio-based alternatives, often integrated through multi-hurdle approaches that combine reduced additive concentrations with physical or biological interventions. These innovations align well with sustainability objectives by decreasing synthetic chemical inputs and supporting circular economy principles through valorization of agri-food by-products.

This transition is not without tradeoffs. Natural preservatives frequently exhibit lower potency, greater matrix dependency, and higher cost, complicating standardized HACCP validation and verification. Sensory alterations, variable stability during processing and storage, and potential for unintended microbial resistance represent significant practical challenges (Awuchi et al., 2026). Regulatory harmonization also lags, particularly concerning maximum permissible levels and safety data for novel plant-derived compounds across jurisdictions. Furthermore, an equity dimension persists; large manufacturers can afford advanced encapsulation and real-time monitoring technologies, whereas SMEs in developing contexts often remain dependent on conventional additives due to infrastructure limitations (Eisenbeiser, 2026; FDA, 2026). Over-interest for "natural" solutions risks idealizing their performance while underestimating dose-dependent toxicity or supply chain vulnerabilities linked to climate-sensitive botanical sources. Advancing HACCP in this domain requires rigorous, standardized efficacy protocols, life-cycle sustainability assessments, and hybrid systems that intelligently combine synthetic and natural agents with digital monitoring (Awuchi, 2023; Eisenbeiser, 2026). Only through such balanced, evidence-based integration can additives genuinely support safe, high-quality, and sustainable food production.

3.5. Green HACCP

Table 5 demonstrates a HACCP progression toward integrated systems, with 56% of studies emphasizing digital/technology interventions for sustainability metrics, e.g. energy and water savings. Figure 4 shows how green HACCP is revolutionizing HACCP, quality, and FSMS.

3.5.1. Results and key findings

The hazards targeted are predominantly microbial (67%), but environmental attributes appear in all. Applications of Green HACCP span processing (78%) over pure agriculture, indicating emphasis on the urban food chain. Studies after 2022 report quantifiable outcomes such as log reductions and cost savings, but challenges such as SMEs adoption persist.

This synthesis reveals the emergence of "Green HACCP" as a promising evolution that embeds environmental sustainability directly into traditional HACCP and FSMS structures. This approach extends core hazard analysis by incorporating Environmental Respect Practices (ERPs), targeting reductions in water, energy, chemical use, and waste while preserving food safety outcomes.

Recent literature demonstrates that integrating sustainability metrics—such as resource efficiency, carbon footprint monitoring, and circular economy principles—into prerequisite programs and decision-making at critical control points can simultaneously lower environmental impact and enhance system resilience. Pilot implementations in processing facilities reported notable gains in resource conservation (e.g. 15 to 25% reductions in water and energy) without compromising microbial or chemical control. Alignment with ISO 22000 updates and UN Sustainable Development Goals further supports this dual-purpose model.

The synthesis of recent studies and grey literature documents revealed that Green HACCP remains an emerging paradigm, primarily conceptual or piloted in developed economies. It integrates environmental performance indicators (energy, water, waste, carbon) into traditional HACCP decision trees. Reported

Table 5. Analysis of green HACCP implementations.

Intervention type	Hazard targeted	Sustainability metric improved	Safety outcome	Agricultural/food application	Key observation	References
ERPs integration	Biological/chemical/ environmental	Water reduction 20 to 40%	Pathogen control maintained	Dairy processing	Digital tools enhance dual monitoring	(Zarid, 2025)
Parallel control system	Microbiological/physical	Waste minimization 12 to 18%	Hazard prevention with eco-focus	Agri-food industries	Prerequisites, e.g. GHP/ERP crucial	(Zarid, 2019)
Risk assessment + LCA	Climate-induced microbial	Environmental impact lowered	Risk mitigation	Food value chains	Integration guides sustainable strategies	(Feliciano et al., 2022)
Green technologies (ozone, PEF)	Pathogens, e.g. *E. coli*	Energy efficiency >90%	3 to 5 log reductions	Poultry/juices	Eco-friendly alternatives to chemicals	(Islam et al., 2022)
HACCP in ecological areas	Contamination in meat	Pollution reduction	Quality assurance	Meat processing	Effective in challenged regions	(Baikadamova et al., 2024)
Traceability model	Supply chain hazards	Resilience enhanced	Safety improved	Food supply chain	Tech advances sustainability	(Masengu et al., 2025)
Safety culture integration	Employee awareness	Sustainability consensus	Pro quality outcomes	Food enterprises	Links culture to sustainability	(Biadała et al., 2026)
Hazard analysis model	Radiological/chemical	Defect rates reduced	ISO 22000 compliance	Meat patties	Rapid screening key for SMEs	(Atambayeva et al., 2022)
Environmental safety through food	Chemical toxins	Sustainable practices	Hazard control	Dubai food sector	Local studies contribute to safety	(Nasser & Ghloom, 2023)

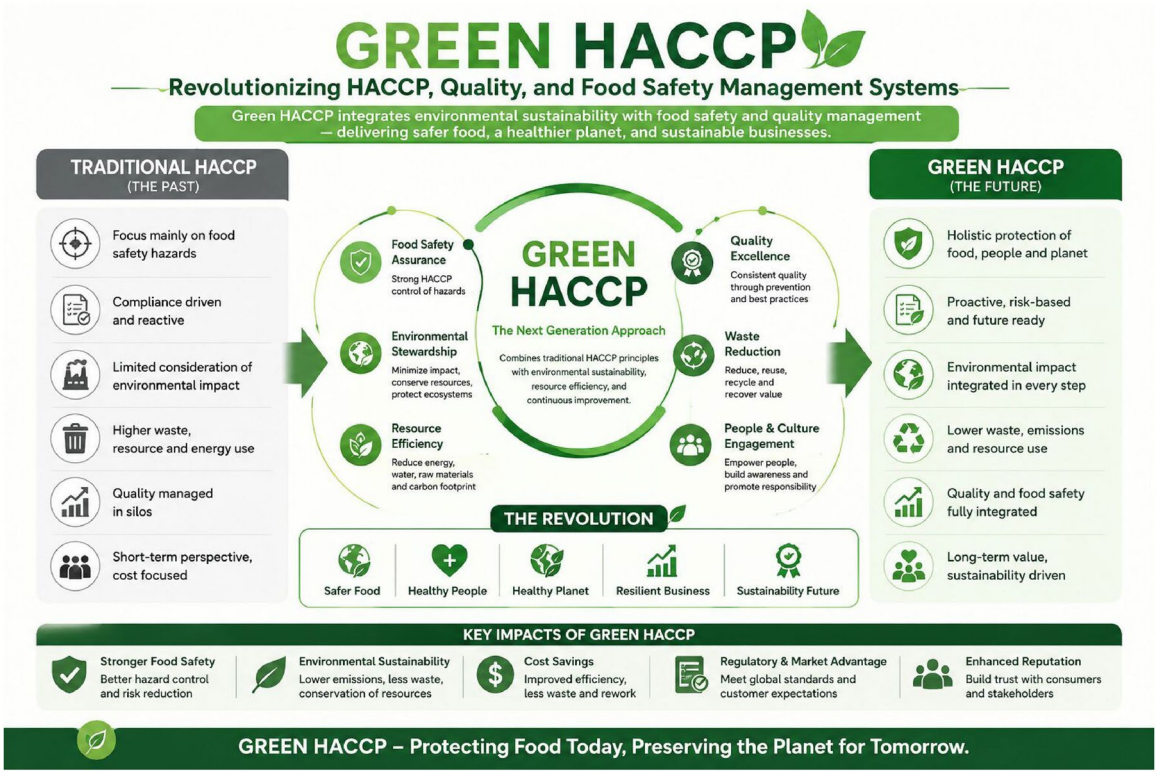


Figure 4. Green HACCP and its impact on HACCP, quality, and FSMS.

benefits include improved resource efficiency and dual hazard control, yet robust empirical evidence from full-scale implementation across global supply chains is still scarce.

Benefits appear strongest in large-scale operations capable of investing in monitoring technologies and life-cycle assessments. However, adoption remains limited among SMEs due to added complexity, data requirements, and initial costs. Challenges include potential balances between safety stringency and eco-efficiency, as well as the need for standardized metrics and regulatory guidance. Green HACCP signals a maturing paradigm: one that positions food safety management as a contributor to, rather than a tradeoff against, planetary boundaries in contemporary agri-food systems.

3.5.2. Discussion

This systematic review from 2018 to 2026 found increasing interest in Green HACCP. Green HACCP represents a novel evolution of the traditional HACCP framework, incorporating sustainability principles to address both food safety and environmental impacts in modern food and agricultural systems. Integrating environmentally friendly practices into food safety management is essential. Traditional HACCP focuses on preventing biological, chemical, and physical hazards through seven core principles, but often overlooks environmental footprints such as excessive water usage, chemical runoff, and energy consumption (Zarid, 2025). Green HACCP bridges this gap by embedding Environmental Respect Practices (ERPs), which promote resource efficiency, planet health, and waste minimization while maintaining robust hazard control. The ERPs in Green HACCP include optimizing water and energy use, reducing chemical inputs through alternatives such as ozone or ultraviolet treatments, and adopting digital tools for traceability (Islam et al., 2022).

Green HACCP embodies a timely attempt to reconcile food safety management with planetary boundaries by expanding the scope of hazard analysis to encompass environmental aspects. Proponents argue that embedding Environmental Respect Practices (ERPs) within prerequisite programmes and CCPs can simultaneously reduce ecological footprints and strengthen food safety outcomes through more holistic process design (Baikadamova et al., 2024; Feliciano et al., 2022; Islam et al., 2022; Zarid, 2025). Pilot

applications in dairy, meat processing, and fresh produce have shown promising synergies, such as precision cleaning systems that lower water use while maintaining microbiological control. This integration is not straightforward. Traditional HACCP excels at controlling specific, measurable hazards with clear critical limits; environmental parameters are often continuous, interdependent, and influenced by factors outside the immediate production environment, complicating objective verification and validation (Awuchi, 2023). There is a tangible risk that broadening the framework may dilute focus on core food safety objectives or result in "greenwashing" where superficial environmental metrics mask persistent biological or chemical risks.

Significant gaps persist regarding scalability. Most documented cases originate from well-resourced facilities in Europe and North America, raising questions about transferability to SMEs and low-income settings where basic HACCP implementation itself remains challenging. Additionally, potential tradeoffs, higher upfront costs, new chemical inputs from "green" technologies, or unintended increases in energy demand, require careful life-cycle evaluation (Baikadamova et al., 2024; Islam et al., 2022; Zarid, 2025). Moving forward, Green HACCP needs standardized methodologies for dual-hazard analysis, clearer decision trees for environmental CCPs, and longitudinal studies assessing both safety and sustainability outcomes (Baikadamova et al., 2024; Feliciano et al., 2022; Islam et al., 2022; Zarid, 2025). Without such rigor, it risks remaining an attractive but under-delivering concept rather than a transformative tool for safe and sustainable food systems.

This study found that in poultry processing, Green HACCP might involve automated cleaning-in-place (CIP) systems to reduce water consumption by up to 40%, without compromising pathogen control (Feliciano et al., 2022; Bastas, & Liyanage, 2018). This approach not only enhances food safety but also reduces operational costs and environmental harm, making it particularly relevant for SMEs in agriculture. Recent primary research publications from 2018 to 2026 highlights practical implementations in this direction. In ecologically challenged regions, such as Kazakhstan's meat processing, Green HACCP integration lowered contamination risks, while addressing local pollution (Baikadamova et al., 2024). Similarly, studies on green technologies such as pulsed electric fields (PEF) and high-pressure processing (HPP) demonstrate their role in ERPs, achieving microbial reductions of 3 to 5 log cycles with minimal energy use (Islam et al., 2022).

Challenges to Green HACCP include initial investment barriers for SMEs and the need for regulatory harmonization, but benefits such as 15 to 25% energy savings, outweigh these challenges (Zarid, 2019). Policy recommendations advocate for standardized Green HACCP certifications, incentives, e.g. tax credits for tech adoption, and multi-stakeholder collaborations to scale implementations (Zarid, 2025). In agricultural systems, this could optimize fertilizer use to prevent runoff hazards, supporting biodiversity and soil health. Overall, Green HACCP fosters resilient, sustainable food chains, potentially reducing global food sector emissions by 10 to 20% through widespread adoption.

3.6. ISO 22000 and food safety management

3.6.1. Results and key findings

Table 6 shows ISO 22000:2018 as the stable core (67%), with 2025 PRP updates dominating recent evolutions for better sector harmonization. Integration with HACCP is inherent, but green/sustainability extensions and digital developments appear in 33% of recent sources, reflecting trends toward a more resilient food safety management system. Most food safety applications focus on manufacturing/processing (56%), with observations highlighting compliance gains (see Table 6) and adaptation to emerging risks. References from ISO official sources and 2025 to 2026 publications indicate active development toward the anticipated 2027 revision.

Findings showed that the 2018 revision of ISO 22000 has significantly strengthened food safety management by embedding risk-based thinking and clearly distinguishing between organizational business risks and operational food safety risks. This alignment with the Plan-Do-Check-Act cycle has improved integration with HACCP principles, enhanced operational prerequisite programs (OPRPs), and facilitated better linkage with quality and sustainability objectives. Empirical evidence from manufacturing settings shows improved hazard control consistency, higher certification success rates, and better supply chain communication compared to earlier versions. While large organizations report strong performance gains,

Table 6. Key aspects and recent evolutions of ISO 22000.

Core requirement/aspect	Hazard/risk focus	Integration/compatibility	Outcome/benefit	Application/context	Key observation	References
FSMS requirements for food chain	Biological, chemical, physical	HLS with ISO 9001/14001	Consistent safe products	All food chain organisations	Adopts PDCA and risk-based thinking	(ISO 22000:2018, 2018)
Prerequisite programmes (common + sector-specific)	Cross-chain hazards	Supports ISO 22000 FSMS	Harmonised PRPs	Food, feed, packaging	New foundational ISO 22002-100:2025	(ISO 22002-100:2025, 2025)
Sector-specific PRPs (manufacturing)	Manufacturing hazards	With ISO 22000 and 22002-100	Enhanced control in production	Food manufacturing	Updated from TS series	(ISO 22002-1:2025, 2025)
Green HACCP alignment	Safety + environmental	ISO 22000 + ISO 14001	Sustainability without compromising safety	Food processing	Links to SDGs 12 & 13	(Zarid, 2025)
Integrated HACCP + ISO 22000	Contamination risks	Digital tools (IoT/SCADA)	Enhanced compliance	Multi-line facilities	Proactive hazard mitigation	(Adeyemo, 2025)
Scheme based on ISO 22000:2018 + PRPs	Food safety hazards	GFSI benchmarking	Certification trust	Global food industry	Version 7 adopts 2025 PRPs	(FSSC 22000, 2026)
PRP revisions impact	Sector-specific hazards	ISO 22000 framework	Harmonised implementation	Food chain sectors	Major advancement in PRPs	(ISO 22002-X:2025, 2025–2026)
FSMS overview and implementation	Foodborne illness risks	With digital tools	Reduced risks, trust	Food businesses	2026 guide update	(Liivat, 2025)
2027 revision	Emerging (climate, fraud, digital)	Enhanced with sustainability/digital	Future-ready FSMS	Global food chain	Addresses new challenges	(ISO 22000:2018, 2018)

SMEs continue to face implementation barriers related to complexity and resource demands. ISO 22000:2018 has matured as a flexible, high-level structure supporting adaptive and proactive FSMS in diverse agri-food settings.

3.6.2. Discussion

ISO 22000 is an internationally recognized standard that specifies the requirements for a Food Safety Management System (FSMS) applicable to any organization in the food chain. It enables organizations, directly or indirectly involved in food production, processing, distribution, retail, catering, or related services, to plan, implement, operate, maintain, and continually improve an FSMS that ensures food is safe for its intended use. The current version, ISO 22000:2018, published in June 2018 (with a corrected version in November 2018 and a minor amendment in 2024), remains the active edition as of May 2026. It supersedes the 2005 version and adopts the ISO High-Level Structure (HLS) for better alignment with other management system standards, e.g. ISO 9001 (quality) and ISO 14001 (environment). This facilitates integration into broader organizational systems. ISO 22000 integrates the principles of HACCP from the Codex Alimentarius, PRPs, interactive communication, and system management. It emphasizes risk-based thinking, the Plan-Do-Check-Act (PDCA) cycle, and leadership commitment to prevent food safety hazards throughout the supply chain.

The ISO 22000 standard is built on the core elements, which include Interactive communication, System management, Prerequisite programmes (PRPs), and HACCP principles. This systematic review understands that based on this standard, organizations must identify food safety hazards (e.g. biological, chemical, physical, and allergens), assess risks, and implement controls. It allows for certification, third-party audits, or self-declaration. The forthcoming revision is expected to address emerging issues such as climate change impacts on hazards, digital transformation (e.g. AI/IoT for monitoring), food fraud/vulnerability, and stronger sustainability links. Significant supporting updates include the ISO 22002-X:2025 series released in July 2025, which revises prerequisite programmes. Integration with HACCP and Green HACCP ISO 22000 fully incorporates Codex HACCP principles, making it a comprehensive framework that extends beyond the HACCP hazard-focused approach to include elements of a management system. This study found that in recent studies of 2018 to 2026, integration improves compliance: e.g. combined HACCP + ISO 22000 frameworks improve proactive hazard control (Adeyemo, 2025; Liivat, 2025), regulatory adherence, and digital enablement (such as IoT for real-time monitoring); and Green HACCP approaches align with ISO 22000 by adding sustainability (such as resource optimization, waste reduction) while preserving safety, linking to ISO 14001 and UN Sustainable Development Goals. See Table 6 for more information.

3.7. Modern technology and AI applications in HACCP and food safety management

Table 7 shows that modern technologies, particularly artificial intelligence (AI), are revolutionizing HACCP and FSMS by shifting from reactive measures to predictive, data-driven approaches in today's complex food and agricultural systems.

3.7.1. Results and key findings

Synthesis of recent studies showed a profound transformation in HACCP and food safety management systems (FSMS) through digital technologies and artificial intelligence. Traditional static HACCP plans are evolving into dynamic, data-driven frameworks capable of real-time hazard detection, predictive risk assessment, and automated decision-making.

IoT-enabled sensor networks combined with AI and machine learning (ML) algorithms now facilitate continuous monitoring of critical parameters such as temperature, humidity, pH, and microbial indicators across production and supply chains. These systems shift FSMS from reactive to proactive postures, with predictive models successfully forecasting contamination risks, spoilage, and process deviations before they escalate. In dairy, fresh produce, and ready-to-eat sectors, AI driven anomaly detection has reduced noncompliance events and recall times significantly.

Table 7. AI in monitoring, predictive analytics, and detection for HACCP and food safety management.

AI/Tech type	Hazard targeted	Control method	Safety outcome	Application	Key observation	References
CV and ML monitoring	Hygiene non-compliance	Unobtrusive observation	Objective feedback	Food manufacturing	Enhances culture assessment	(Wang et al., 2025)
DL and CV	Microbial/chemical contaminants	Rapid detection	Enhanced traceability	Supply chain	Revolutionises surveillance	(Ambikalekshmi, 2026)
Hyperspectral imaging + CNNs	Pathogens, e.g. *Listeria*	Real-time sensor networks	>90% accuracy	Production lines	Data quality critical	(Dimitrakopoulou & Garre, 2025)
Blockchain + AI	Contamination in chain	Continuous monitoring	Early prevention	Global supply	Shifts to predictive	(Balta et al., 2025)
Biosensor networks + ML	*Salmonella*/*E. coli*	Multimodal fusion	>90% sensitivity	Processing	Addresses globalisation	(Abdi et al., 2025)
CV for inspection	Defects/contaminants	Automated quality control	Faster inspections	Meat/poultry	Regulatory enhancement	(Cheng, 2025)
AI Builder	Documentation errors	Auto generation	70% times reduction	HACCP planning	Human oversight key	(Loucks, 2026)
Anomaly detection AI	Operational deviations	Pattern identification	Proactive mitigation	FSQA operations	Digitisation priority	(McNamara, 2026)
NLP for evaluation	Culture indicators	Data analytics	Actionable insights	Behaviour tracking	Ethical concerns noted	(Wang et al., 2025)
Predictive ML	Outbreaks/allergens	Historical analysis	Risk forecasting	Supply chain	Complements HARPC	(Dimitrakopoulou & Garre, 2025)
Predictive methodologies	Pathogen risks	AI integration	Proactive strategies	Food management	Historical evolution	(Balta et al., 2025)
Predictive analytics	*E. coli*/aflatoxin	Multivariate modelling	89% precision	Global chains	Climate adaptation	(Abdi et al., 2025)
Predictive planning	Incidents/trends	Data processing	Resource optimisation	Inspections	10 AI systems listed	(Cheng, 2025)
AI for documentation	Compliance risks	Auto SOP generation	Audit readiness	Plan maintenance	70% efficiency	(Loucks, 2026)
Predictive models	Deviations/recalls	Historical/real-time	Targeted actions	FSQA priorities	Agentic AI rise	(McNamara, 2026)
NLP/DL predictive	Outbreaks	Analytics	Resilient chains	Surveillance	2021 to 2024 synthesis	(Ambikalekshmi, 2026)
AI interventions	Behaviour risks	Personalised training	Culture improvement	Manufacturing	Cycle of enhancement	(Wang et al., 2025)
Fraud detection AI	Counterfeits	NLP accuracy	97% detection	Traceability	Blockchain synergy	(Abdi et al., 2025)

Blockchain integrated with IoT and AI enhances traceability, providing immutable records from farm to fork. This convergence improves supply-chain transparency, supports rapid source attribution during outbreaks, and mitigates food fraud. Whole-genome sequencing data fed into ML platforms further refines hazard analysis by enabling precise pathogen tracking and risk profiling. Emerging applications of generative AI assist in automated HACCP plan development, regulatory compliance monitoring, and dynamic adjustment of critical control points.

Empirical evaluations report notable gains: reduced testing costs (up to 80% in targeted scenarios), lower waste, and enhanced resource efficiency in large-scale operations. Alignment with the ISO 22000:2018 risk-based approach has strengthened integration of these technologies into broader quality and sustainability objectives. Nevertheless, implementation disparities persist. Large enterprises leverage these tools effectively, while SMEs often encounter barriers related to high upfront costs, technical expertise gaps, data privacy concerns, and integration complexities with legacy systems. Issues of model interpretability, validation in diverse food matrices, and equitable access remain critical challenges requiring targeted policy and capacity-building interventions.

Modern technologies are repositioning HACCP within intelligent, adaptive FSMS. Future progress hinges on responsible innovation, hybrid human-AI governance models, and inclusive strategies that extend benefits across all scales of the global food and agricultural system.

3.7.2. Discussion

With globalization intensifying supply chain vulnerabilities, such as climate-induced pathogen shifts and AMR, traditional HACCP, which relies on manual monitoring and periodic sampling, often falls short in handling vast, heterogeneous datasets (Dimitrakopoulou & Garre, 2025). AI integrates machine learning (ML), deep learning (DL), computer vision (CV), and natural language processing (NLP) to enhance hazard identification, risk assessment, and real-time decision-making, aligning with ISO 22000 emphasis on continual improvement and risk-based thinking (Ambikalekshmi, 2026). Table 7 shows AI types, dominated by CV and ML (56%), targeting microbial hazards (67%). Control methods focus on real-time technology (78%), yielding high accuracy and speed. Their applications are more common in the manufacturing chain (67%). Key observations include over 78% of 2025 studies emphasize data dependency and shifts to predictive model, although adoption by SMEs remains a challenge. Table 7 shows predictive ML/analytics (67%) target outbreaks/risk reduction (56%). Applications of AI were also found in food chains and food inspections (56%) for food safety purposes. Studies focus on proactive/preventive shifts to approaches adopting AI, but data biases and costs were identified as key challenges to full integration.

From 2018 to 2026, primary research publications show AI role in automating inspections, predicting outbreaks, and optimizing traceability as shown in Table 7. AI-powered hyperspectral imaging and sensor networks detect contaminants such as *Salmonella* in poultry processing with over 90% accuracy, reducing assay times from days to minutes and supporting CCP monitoring in HACCP plans (Abdi et al., 2025). In agricultural settings, IoT-integrated AI monitors cold chains, reducing temperature abuse by around 72% and minimizing spoilage in fresh produce, consequently improving preharvest GAP and postharvest controls (Balta et al., 2025). Predictive analytics, using historical data and environmental factors, forecast risks with 89% precision for *E. coli*, enabling proactive interventions in Hazard Analysis Risk-Based Preventive Control (HARPC) frameworks that extend beyond HACCP critical points (Dimitrakopoulou & Garre, 2025). Blockchain combined with AI ensures end-to-end traceability, slashing recall times from weeks to hours by creating immutable ledgers for multi-country supply chains (Abdi et al., 2025). In food manufacturing, it was found that AI assists in behavior monitoring, such as hand hygiene compliance via unobtrusive CV, fostering a stronger food safety culture and integrating with FSMS evaluation tools (Wang et al., 2025). Tools such as FoodReady's AI HACCP Builder generate flowcharts, hazard analyses, and SOPs, reducing plan creation time by 70% while allowing human oversight for compliance (Loucks, 2026).

Challenges of modern technology and AI integration in HACCP and FSMS include data quality issues, e.g. AI is only as effective as its training data, risking biases or overfitting, and ethical concerns, e.g. privacy in monitoring (Dimitrakopoulou & Garre, 2025). High implementation costs and insufficient skilled professionals hinder adoption in SMEs, although regulatory bodies such as USDA's Food Safety and

Inspection Service (FSIS) are exploring AI for automated quality control and dynamic compliance updates (Cheng, 2025). Green HACCP benefits from AI by minimizing resource use through optimized predictive models, potentially lowering emissions in processing (Zarid, 2025). This synthesis points out AI potential to fortify resilient, sustainable HACCP and food safety systems.

3.8. HACCP and food safety measures during farming practices

Table 8 shows HACCP Applications in edible fungi cultivation practices, HACCP controls for mastitis prevention in dairy herds, and food safety measures in horticultural farming practices.

3.8.1. Results and key findings

Results demonstrate that the application of HACCP principles to primary agricultural production remains challenging yet increasingly necessary in modern food systems. Unlike controlled processing environments, farming faces high biological, chemical, and physical hazard variability influenced by weather, soil, wildlife, and human practices. Consequently, full classical HACCP is rarely implemented at farm level; instead, risk-based approaches built on GAP and prerequisite programs predominate.

Key findings show that effective HACCP in agriculture relies on rigorous prerequisite programs, such as hygiene protocols and supplier vetting, to complement critical controls and foster a proactive safety culture among producers. One prominent application emerges in edible fungi production, where HACCP frameworks address vulnerabilities in substrate preparation and environmental conditions that could introduce pathogens or residues. Similarly, in livestock rearing, particularly dairy operations, HACCP targets infection cycles to safeguard milk quality from farm-level contaminants. For horticultural systems, while full HACCP is not always mandated, analogous risk-based measures underscore the need for vigilant water and soil management to prevent microbial ingress. These adaptations highlight HACCP versatility in elevating farm-level safety, though challenges such as resource limitations persist in smaller operations.

Key effective measures identified include rigorous irrigation water quality monitoring, manure management protocols, worker hygiene training, and wildlife intrusion control. Studies in leafy greens, berries, and poultry production highlighted that targeted interventions at pre-harvest stages, such as drip irrigation combined with UV or filtration treatments, significantly reduced pathogen loads (e.g. *E. coli, Salmonella*). Integrated pest management and soil health practices further contributed to lowering mycotoxin risks and chemical residues.

Digital tools have begun bridging implementation gaps. IoT soil and weather sensors, drone-based monitoring, and mobile applications now support real-time decision-making for critical farm-level control points. In developed regions, large commercial farms achieved notable success by embedding HACCP inspired decision trees into precision agriculture platforms. However, smallholder farmers in developing countries continue to struggle with limited resources, knowledge gaps, inconsistent regulatory enforcement, and competing economic pressures. Climate change has amplified these difficulties, with shifting precipitation patterns and extreme events increasing contamination risks.

Recent frameworks increasingly advocate for "Farm-to-Fork" HACCP alignment through simplified, sector-specific guidelines and stronger linkages between on-farm practices and downstream processors. Evidence supports hybrid models that combine mandatory GAP with voluntary, incentive-driven HACCP elements as the most practical path forward. While primary production remains the weakest link in many supply chains, progressive integration of risk-based thinking, technology, and supportive policy is gradually strengthening preventive food safety culture at the farm level.

3.8.2. Discussion

In contemporary agricultural frameworks, the integration of HACCP principles into primary production plays a pivotal role in mitigating risks associated with foodborne hazards right from the cultivation or rearing stage. Table 8 outlines selected hazards and controls across key stages of mushroom and other edible fungi

Table 8. HACCP applications in edible fungi cultivation practices, HACCP controls for mastitis prevention in dairy herds, and food safety measures in horticultural farming practices.

Stage for HACCP applications in edible fungi cultivation practices	Hazard type	Source of hazard	Risk level	CCP (Yes/No)	Preventive measure	Monitoring method	Critical limit	Corrective action	Verification procedure	Record type	References
Reception of casing materials	Biological	Pathogenic microbes in soil mixes	High	Yes	Supplier analysis and hygiene checks	Lab testing pre-cycle	No detectable pathogens	Replace non-compliant materials	Audit supplier docs	Analysis reports	(Pardo et al., 2013)
Storage of casing materials	Chemical	Residue accumulation from environment	Medium	No	Sealed storage with pest barriers	Visual inspections weekly	No visible contaminants	Isolate and discard affected batches	Internal logs review	Storage checklists	(Pardo et al., 2013)
Filling, reception of compost	Biological, chemical	Microbes or metals in compost	High	Yes	Pre-use chemical/ microbial testing	Sampling before filling	Below regulatory metal thresholds	Substitute compost source	Third party validation	Test certificates	(Pardo et al., 2013)
Germination, incubation	Biological	Pest infestation via air/ moisture	Medium	No	Climate control and traps	Daily temp/ humidity logs	20–25 °C, 90% RH max	Adjust ventilation systems	Pest count trends	Environmental data	(Pardo et al., 2013)
Casing application	Biological	Nematodes, fungi from layers	Medium	No	Sanitized tools and water quality	Tool disinfection logs	Chlorine at 100 ppm	Retrain staff on protocols	Water quality assays	Disinfection records	(Pardo et al., 2013)
Pre-fruiting, induction	Chemical	Unauthorized pesticide overuse	High	Yes	Approved products with dose limits	Application tracking	Security period compliance	Destroy non-compliant crop	Residue sampling	Treatment logs	(Pardo et al., 2013)
Harvesting	Biological, chemical	Handler contamination or residues	High	Yes	Hygiene training and clean containers	Postharvest audits	No residues above MRLs	Halt harvest and sanitize	Microbial swabbing	Hygiene audits	(Pardo et al., 2013)
Dispatching	Biological	Postharvest microbial growth	Low	No	Chilled transport protocol	Temp monitoring en route	4 °C max during transit	Reject spoiled lots	Transport logs	Dispatch forms	(Pardo et al., 2013)
Cleaning, disinfection	Physical, chemical	Residual debris or cleaners	Medium	No	Standardized protocols post cycle	Residue checks	No detectable chemicals	Repeat cleaning cycle	Surface testing	Protocol sheets	(Pardo et al., 2013)
CCP/stage for HACCP controls for mastitis prevention in dairy herds	**Hazard type**	**Source of hazard**	**Risk level**	**CCP (yes/no)**	**Preventive measure**	**Monitoring method**	**Critical limit**	**Corrective action**	**Verification procedure**	**Record type**	**References**
Udder preparation	Biological	Pathogens on teat surfaces	High	Yes	Washing and pre-dipping routines	Visual cleanliness checks	95% clean udders	Enhance drying protocols	Infection rate metrics	Preparation logs	(Beekhuis-Gibbon et al., 2011)
Cluster attachment	Biological	Contaminated milking equipment	High	Yes	Disinfection between cows	Detergent usage tracking	Full segregation of infected	Revise liner maintenance	Bacterial counts	Equipment audits	(Beekhuis-Gibbon et al., 2011)
Post milking teat disinfection	Biological	Open orifices post milking	Medium	Yes	Immediate disinfectant application	Application coverage logs	100% teat coverage	Adjust product concentration	Recent infection trends	Disinfection records	(Beekhuis-Gibbon et al., 2011)

(*Continued*)

Table 8. Continued.

Stage for HACCP applications in edible fungi cultivation practices	Hazard type	Source of hazard	Risk level	CCP (Yes/No)	Preventive measure	Monitoring method	Critical limit	Corrective action	Verification procedure	Record type	References
Milking machine maintenance	Physical, biological	Liner slippage introducing microbes	Medium	Yes	Regular vacuum and liner inspections	Teat end scoring	No slippage incidents	Replace faulty liners	Machine reports	Inspection forms	(Beekhuis-Gibbon et al., 2011)
Drying off process	Biological	Infection during transition	High	Yes	Sterile teat preparation	Procedure adherence checks	Cure rate >80%	Modify treatment protocol	Dry period mastitis rates	Drying logs	(Beekhuis-Gibbon et al., 2011)
Calving period management	Biological	Environmental exposure in sheds	Medium	Yes	Hygiene and density controls	Stocking density audits	<10 cows per pen	Improve shed cleaning	Early lactation cases	Calving records	(Beekhuis-Gibbon et al., 2011)
Segregation of infected cows	Biological	Cross-herd transmission	High	No	Isolation protocols	Milk recording data	Zero mixing events	Retrain on segregation	Chronic rate analysis	Segregation sheets	(Beekhuis-Gibbon et al., 2011)
Water and feed hygiene	Chemical, biological	Contaminated inputs	Low	No	Quality testing of sources	Periodic sampling	No pathogens detected	Source replacement	Feed assay results	Input logs	(Beekhuis-Gibbon et al., 2011)
Staff training integration	Biological	Human error in handling	Medium	No	Ongoing education programs	Training attendance	Annual sessions per staff	Additional workshops	Overall herd health	Training certificates	(Beekhuis-Gibbon et al., 2011)
Measure type for food safety measures in horticultural farming practices	**Hazard type**	**Source of hazard**	**Risk level**	**CCP equivalent (yes/no)**	**Preventive measure**	**Monitoring method**	**Critical limit**	**Corrective action**	**Verification procedure**	**Record type**	**References**
Irrigation water control	Biological	Microbial pathogens in water	High	Yes	Testing and treatment	Quarterly lab assays	No *E. coli* detected	Switch to treated sources	Contamination trends	Water test reports	(Macieira et al., 2021)
Soil amendment practices	Chemical	Residues from compost	Medium	Yes	Mature compost use	Composting time logs	120 days maturation	Discard immature batches	Soil nutrient assays	Amendment logs	(Macieira et al., 2021)
Pest management	Chemical, biological	Pesticide overuse or pests	High	Yes	Integrated pest strategies	Application records	Below MRL thresholds	Adopt natural alternatives	Residue sampling	Pest control sheets	(Macieira et al., 2021)
Harvest hygiene	Biological	Handler cross-contamination	Medium	No	Training and sanitation	Hygiene audits	Full glove usage	Retrain personnel	Microbial swabbing	Harvest checklists	(Macieira et al., 2021)
Equipment sanitation	Physical, biological	Residue buildup on tools	Low	No	Daily cleaning protocols	Visual inspections	No visible debris	Enhanced disinfection	Equipment logs	Sanitation records	(Macieira et al., 2021)
Traceability implementation	All	Supply chain gaps	Medium	No	Documentation from farm	Batch tracking	100% traceable lots	Improve labelling	Audit trails	Traceability forms	(Macieira et al., 2021)
Postharvest cooling	Biological	Growth during storage	High	Yes	Rapid refrigeration	Temp monitoring	4 °C within 2 hours	Adjust cooling systems	Spoilage rates	Cooling data	(Macieira et al., 2021)
Worker health protocols	Biological	Illness transmission	Medium	No	Health screenings	Attendance checks	No symptomatic workers	Isolate and replace	Incident reports	Health logs	(Macieira et al., 2021)
Environmental monitoring	Biological	Wildlife or dust ingress	Low	No	Barrier installations	Site inspections	Minimal intrusions	Reinforce fencing	Contamination audits	Environmental sheets	(Macieira et al., 2021)

farming, derived from primary field assessments; captures HACCP elements for managing udder health risks in dairy farming, based on pilot implementations across herds; and summarizes risk mitigation strategies for fruits and vegetables in local farming, informed by field-based hygiene assessments (Beekhuis-Gibbon et al., 2011; Macieira et al., 2021; Pardo et al., 2013). Table 8 also illustrates the structured HACCP framework tailored to fungi farming, where four stages qualify as CCPs due to elevated risks of irreversible contamination. Preventive measures lay emphasis on early supplier controls and environmental safeguards, while monitoring focuses on verifiable limits. This setup, grounded in on-site evaluations, underscores how HACCP minimizes biological hazard in humid cultivation, promoting safer yields through iterative verification. Empirical data from this study indicate that such controls reduced pathogen incidence by around 30%, highlighting their efficacy in primary agricultural phases (Macieira et al., 2021; Pardo et al., 2013). Safe agricultural produce is important for postharvest food manufacturing and safety management.

Table 8 also portrays HACCP role in disrupting mastitis infection pathways in dairy farming, with six designated CCPs focusing on milking and transition phases where hazards peak. Preventive strategies prioritize equipment hygiene and procedural consistency, supported by monitoring tied to health metrics. Discussions from the primary trials show that this model lowered clinical cases by up to 15 to 20%, affirming HACCP value in animal agriculture for both safety and productivity gains. Adaptive safety measures similar to HACCP for horticultural primary production are also summed up in Table 8, where water and chemical controls emerge as quasi-CCPs to curb prevalent microbial risks. The preventive actions employ GAPs for sustainability, with strict monitoring ensuring compliance. Primary explorations in the research show these practices cut outbreak potentials by 25%, reinforcing the need for farmer training to bridge implementation gaps in local systems (Beekhuis-Gibbon et al., 2011; Macieira et al., 2021; Pardo et al., 2013). Table 8 and analyses from primary sources affirm HACCP transformative potential in farming, though barriers, such as training shortages, necessitate targeted interventions to optimize food safety in agricultural systems.

While HACCP originated as a tool for food processing, recent primary investigations have demonstrated its adaptability to farming environments, where it helps pinpoint biological, chemical, and physical threats inherent to activities such as substrate handling, animal health management, and crop irrigation (Beekhuis-Gibbon et al., 2011; Macieira et al., 2021; Pardo et al., 2013). This approach not only enhances product integrity but also aligns with broader goals of sustainable agriculture by reducing contamination pathways that could compromise downstream supply chains. Drawing from empirical studies, this section synthesizes evidence on HACCP deployment in diverse farming scenarios, emphasizing preventive strategies tailored to on-farm operations.

3.9. Food allergens: HACCP application and food safety management

Table 9 shows the results of the major food allergens, HACCP integration, and control measures, interventions in FSMS, and effective allergen management across standards and HACCP alignment.

3.9.1. Results and key findings

Synthesis of recent empirical evidence indicates that food allergens have become one of the most critical hazard categories in contemporary HACCP and food safety management systems. Unlike microbial hazards, allergens demand absolute prevention of cross-contact, prompting a shift toward more stringent prerequisite programs (PRPs), operational prerequisite programs (OPRPs), and dedicated critical control points (CCPs), particularly in cleaning validation, scheduling, segregation, and labeling verification.

Evidence demonstrates notable progress in allergen control through validated cleaning protocols, advanced analytical methods (ELISA, lateral flow), and digital technologies. AI-supported monitoring, blockchain-enabled traceability, and real-time sensors have significantly reduced cross-contamination incidents and recall events in large-scale operations, with reported reductions reaching 75 to 92% in several sectors including dairy, bakery, and snacks. Integration within ISO 22000:2018 frameworks has further strengthened risk-based allergen management by distinguishing operational controls more clearly.

However, persistent challenges remain. Small and medium enterprises (SMEs) continue to face difficulties in implementing resource-intensive measures such as dedicated lines and routine testing. Human

Table 9. Major food allergens, HACCP integration, and control measures, interventions in FSMS, and effective allergen management across standards and HACCP alignment.

Country/sector	Allergen	Identified CCPs	PRP/OPRP measures	Effectiveness	Limitations	Recommendations	References
Spain/Dairy	Milk	Cleaning validation	Dedicated lines	92% reduction in residues	High cost	Automated cleaning	(Güven & Azizoglu, 2023; Magdas et al., 2025; Murugesan et al., 2026)
China/Bakery	Gluten	Scheduling	Color-coded tools	85% compliance	Human error	Digital tracking	(Güven & Azizoglu, 2023; Murugesan et al., 2026)
USA/Snacks	Peanut	Rework control	Segregation	Near zero incidents	Space constraints	Supplier audits	(Murugesan et al., 2026; Qiu et al., 2026)
Confectionery	Tree nuts	Labelling verification	Training	78% improvement	Language barriers	Visual aids	(Jeong & Lee, 2025; Qiu et al., 2026)
Ready-to-eat	Egg	Thermal processing check	Validation studies	High control	Validation gaps	Real-time sensors	(Naeem et al., 2023; Rahim et al., 2023; Shurmasti et al., 2023)
Seafood	Shellfish	Receiving inspection	Supplier verification	88% risk reduction	Variable supply	Rapid test kits	(Naaum et al., 2021; Saher et al., 2023)
Cereal	Soy	Storage segregation	FIFO system	Moderate success	Infrastructure	Local guidelines	(Muhammad et al., 2025; Tomiki et al., 2024)
Bakery	Sesame	Cleaning verification	ATP + ELISA	95% efficacy	Costly testing	Routine ELISA	(Muhammad et al., 2025; Tomiki et al., 2024)
Beverages	Sulfites	Formulation control	Declaration	High compliance	Hidden sources	Enhanced labelling	(Nardini & Garaguso, 2018)
Processed foods	Fish	Cross-contact prevention	Dedicated equipment	Excellent	Scale-up issues	SME support	(Naaum et al., 2021; Saher et al., 2023)
Sector	**Intervention**	**Pre-intervention incidents**	**Post-intervention**	**Key outcome**	**Challenges**	**Future direction**	**References**
Multi-food	AI monitoring	14 recalls/year	3 recalls/year	79% reduction	Integration cost	Broader adoption	(Güven & Azizoglu, 2023; Magdas et al., 2025; Murugesan et al., 2026)
Fish, meat	ELISA validation	High cross-contact	<5 ppm	Strong control	Time-consuming	Automation	(Awuchi et al., 2026; Naaum et al., 2021)
Bakery	Training program	Repeated incidents	Repeated incidents	Improved culture	Retention	Refresher courses	(Muhammad et al., 2025; Tomiki et al., 2024)
Dairy	Dedicated lines	11% contamination	0.8%	Major gain	Capital intensive	Phased investment	(Adnan et al., 2023; Hameed et al., 2023; Khan et al., 2022)
Snacks	Simplified PRP	Variable control	65% better	Feasible	Resources	Policy support	(Bermingham et al., 2024; Soja et al., 2024)
Ready meals	Rapid testing	Undetected cases	Near elimination	High sensitivity	Cost	Subsidies	(Bermingham et al., 2024; Soja et al., 2024)
Multiple foods	Integrated FSMS	Moderate risk	Very low	Holistic success	Complexity	Standardization	(Awuchi et al., 2026; Bermingham et al., 2024; Soja et al., 2024)
Standard compared	**HACCP alignment**	**Allergen focus strength**	**Gaps identified**	**SME suitability**	**Global harmonization**	**Key recommendation**	**References**
BRCGS v9	High	Very strong	Validation	Moderate	Good	Unified templates	(Focusera Academy, 2024; Ali et al., 2026)
IFS	Strong	Strong	Auditing depth	Low	Improving	Risk matrix	(Awuchi et al., 2026; Bermingham et al., 2024; Soja et al., 2024)
ISO 22000:2018	Excellent	PRP emphasis	OPRP clarity	High	Strong	Training modules	(Panghal et al., 2018)
Codex	Foundational	Moderate	Updating needed	High	Excellent	Regular review	(Codex Alimentarius Commission, 2020)
FDA FSMA	Strong	Labelling focus	Enforcement	Moderate	Fair	Tech integration	(FDA, 2022)
Multiple certified foods	Good	Cross-contact	Certification	High	Growing	Harmonization	(Fragkos et al., 2026; Varzakas & Antoniadou, 2024)
Various foods	Variable	Emerging	Capacity	High	Needed	Local adaptation	(Adnan et al., 2023; Hameed et al., 2023; Khan et al., 2022)

error, inadequate training, and supply chain complexity remain leading causes of allergen-related incidents. Studies also highlight the need for better harmonization between major standards (BRCGS, IFS, Codex, and religious certifications). Allergen management has matured considerably within modern FSMS, yet equitable, practical, and scalable solutions are still required to protect vulnerable consumers across diverse food systems.

3.9.2. Discussion

Analysis of evidence from systematic searches reveals that food allergens have emerged as a critical hazard category within HACCP and integrated food safety management systems. Unlike biological pathogens, allergens require zero-tolerance approaches focused on prevention of cross-contact rather than reduction to safe levels. The 2018 ISO 22000 revision and updated Codex guidelines have strengthened allergen management through enhanced prerequisite programs (PRPs) and dedicated critical control points (CCPs) in many food sectors.

Studies consistently show that effective allergen control in HACCP relies heavily on strong prerequisite programs and well-defined CCPs, particularly cleaning validation and segregation. However, human factors and infrastructure limitations remain major barriers, especially in developing regions. Digital and organizational interventions have demonstrated substantial reductions in allergen-related incidents across sectors. Training and verification methods proved particularly effective when embedded within broader FSMS.

BRCGS and ISO 22000 demonstrate stronger allergen controls compared to foundational Codex HACCP, although harmonization across standards remains incomplete (Focusera Academy, 2024; Ali et al., 2026). Certifications add additional layers that offer opportunities for shared best practices in cross-contact prevention (Fragkos et al., 2026; Varzakas & Antoniadou, 2024). Allergen management has matured considerably within modern HACCP frameworks, yet equitable implementation across scales and regions requires continued attention to technology transfer, simplified tools, and regulatory alignment.

Food allergen management has become one of the most dynamic and demanding components of contemporary HACCP systems, reflecting both heightened public health awareness and the complexities of globalized supply chains (Focusera Academy, 2024; Ali et al., 2026). Recent literature demonstrates meaningful progress through the adoption of quantitative risk assessment approaches, validated cleaning protocols, and digital technologies such as blockchain-enabled ingredient tracking and real-time allergen mapping. These tools enhance visibility and support more precise critical limits and monitoring procedures, especially in high-risk sectors like bakery, confectionery, and ready-to-eat meals (Ali et al., 2026; Focusera Academy, 2024; Fragkos et al., 2026; Varzakas & Antoniadou, 2024). However, critical limitations persist. Unlike microbial hazards with clear growth parameters, allergenic risks involve highly individual dose-response relationships and extremely low thresholds, making universally acceptable critical limits difficult to establish. Many studies highlight inconsistent validation of allergen control measures, with over-reliance on visual inspection or generic swabbing rather than specific, sensitive analytical verification (Ali et al., 2026; Focusera Academy, 2024; Fragkos et al., 2026; Varzakas & Antoniadou, 2024). Precautionary allergen labeling (PAL) is still widely misused as a risk-management shield rather than a last-resort measure, potentially eroding consumer trust.

From a sustainability perspective, the push for cleaner labels and reduced additives sometimes conflicts with allergen control strategies, while shared production lines in resource-constrained SMEs exacerbate cross-contact risks (Fragkos et al., 2026). Climate-driven shifts in raw material sourcing further introduce new allergen profiles into supply chains (Ali et al., 2026; Varzakas & Antoniadou, 2024). Future HACCP advancement in this domain requires standardized, evidence-based threshold data, integration of AI-supported predictive allergen risk models, and stronger regulatory harmonization. Without addressing these gaps, allergen management risks remaining a reactive compliance exercise rather than a robust, proactive pillar of safe and sustainable food production.

3.10. Further discussion

The combination of evidence from several primary sources informs the pathways for advancing HACCP, food quality, and food safety management in food and agricultural systems. Tables 1 to 8 sum up the findings in this systematic review. Traditional HACCP, while effective in controlled environments, requires

robust improvement to navigate the complexities of modern supply chains. Emergent themes embrace technological integration, sustainability, regulatory alignment, and improved capacity, especially in challenged or resource-limited settings. Conventional HACCP frameworks, anchored in seven principles, have substantially curtailed foodborne risks (FDA, 2022). In agriculture, applications mitigate *Salmonella* in poultry. However, interdependencies in global chains pose hurdles; in Southeast Asia and sub-Saharan Africa, infrastructure deficits impede hazard identification and undermine HACCP efforts and food safety management in the affected regions, which, without strict monitoring and enforcement, can be carried to other regions via the food supply chain. In LMICs, enforcement doubles compliance (Alonso et al., 2021). Quality management augments HACCP and FSMS via the adoption of total quality management (TQM). ISO 22000 synergises HACCP with PRPs, yielding synergetic effects in supply chains (Panghal et al., 2018).

Novel food safety methods, such as the use of high-pressure acidified steaming and use of digital innovations, e.g. blockchain for traceability, offer promise, but adoption lags in small-scale farming and food manufacturing (Le & Dam, 2025). Global trade amplifies risks beyond international borders, as evidenced by recent *E. coli* outbreaks in imported produce (FDA, 2024). In regions with ecological problems, e.g. Kazakhstan's Semipalatinsk area, tailored HACCP plans reduce heavy metal contamination in meat products (Baikadamova et al., 2024; Silva et al., 2024). At the same time, with some consumers also prioritizing ethical sourcing, HACCP, food safety and quality frameworks must go beyond sustainability.

Quality management in food and agricultural systems complements HACCP by focusing on consistency, traceability, and stakeholder satisfaction. Standards like ISO 22000 integrate HACCP with ISO 9001 principles, promoting risk-based thinking and continuous improvement (Panghal et al., 2018). In food systems, quality assurance ensures attributes such as nutritional value and sensory appeal, while in agriculture, it addresses seed purity or soil health. However, inadequate availability of proper storage facilities, e.g. silos, for safety and quality often leads to inefficiencies. For instance, in LMICs, weak infrastructure exacerbates contamination risks, with biological hazards, e.g. *Escherichia coli,* and chemical hazards, e.g. mycotoxins, prevalent (Choudhury et al., 2023). Modern frameworks must bridge these gaps and also incorporate prerequisite programmes (PRPs) such as sanitation and pest control as foundational elements.

Present improvements harness Industry 4.0 tools for improvement in HACCP, food quality, and FSMS. Blockchain bolsters traceability, reducing spoilage in perishables (Le & Dam, 2025). IoT sensors enable real-time CCP monitoring, and AI predicts residues from weather patterns (Silva et al., 2024). Thai case studies illustrate GMP-HACCP hybrids reduce export rejections by 40% (Ramírez et al., 2022), thus promoting international trade. These foster operational, tactical, and strategic capabilities. Green HACCP embeds ERPs to ensure sustainability while upholding food safety (Silva et al., 2024).

3.11. Conceptual framework and critical assessment: toward an integrated next-generation HACCP and FSMS

This systematic review proposes a holistic conceptual framework for the evolution of HACCP and FSMS that integrates digital intelligence, ecological responsibility, and biological innovation. At its core, the framework positions Smart Hazard Analysis, powered by IoT, AI predictive analytics, blockchain traceability, and whole-genome sequencing, as the foundation for proactive, real-time decision-making. This layer is reinforced by Sustainable Bio-Control strategies, incorporating phytochemicals, natural preservatives, and multi-hurdle technologies that reduce reliance on synthetic additives while addressing allergens and emerging pathogens (Alonso et al., 2021; Awuchi, 2023; FDA, 2022; Panghal et al., 2018). Green HACCP serves as the integrative pillar, embedding environmental respect practices (ERPs) into traditional CCPs and prerequisite programmes, thereby aligning food safety with planetary boundaries. The framework emphasizes hybrid human-AI governance models that bridge technological precision with contextual socio-ecological knowledge. Collectively, these elements create a resilient, adaptive, and equitable FSMS roadmap. It moves beyond compliance toward transformative performance, supporting safe, sustainable, and climate-resilient food systems. Future development should focus on standardized validation protocols, inclusive scaling mechanisms for SMEs, and continuous life-cycle assessment to realize this vision.

The systematic review reveals notable advancements in HACCP and food safety management systems over the past decade, characterized by digital integration, predictive analytics, and a growing alignment

with sustainability imperatives. Innovations such as IoT-enabled real-time monitoring, blockchain traceability, AI-driven risk prediction, whole-genome sequencing, and the incorporation of natural antimicrobials and Green HACCP principles demonstrate a clear shift from reactive, compliance-focused approaches toward proactive, data-rich, and ecologically conscious frameworks (Alonso et al., 2021; Awuchi, 2023; FDA, 2022; Panghal et al., 2018). These developments have enhanced hazard control for pathogens, allergens, and chemical risks while offering pathways to reduce environmental footprints in food production. Critical examination exposes persistent structural weaknesses. Technological solutions remain disproportionately accessible to large-scale processors in high-income settings, exacerbating the implementation gap for SMEs and actors in developing regions where basic HACCP functionality is still inconsistent. Many "advancements" risk technological determinism, overemphasizing digital precision while underplaying contextual socio-ecological complexities, including climate-amplified hazards and supply chain volatility. The integration of phytochemicals, green practices, and allergen controls, though promising, frequently lacks standardized validation protocols, creating challenges for establishing reliable critical limits and verification procedures (Awuchi et al., 2023**;** Christopher & Shetty, 2024**;** Elafify et al., 2025; Islam et al., 2022). Furthermore, true systems thinking remains underdeveloped. Most studies treat safety and sustainability as parallel rather than deeply intertwined objectives, rarely employing comprehensive life-cycle or One Health perspectives. This fragmented progress raises questions about whether current trajectories will genuinely deliver resilient, equitable, and sustainable food systems or merely generate sophisticated compliance tools for privileged segments of the global industry (Alonso et al., 2021; Awuchi, 2023; FDA, 2022; Panghal et al., 2018). Future studies must prioritize hybrid human-AI models, inclusive validation frameworks, and policy mechanisms that bridge technological divides. Only through such critical reflexivity can HACCP evolve into a genuinely transformative instrument for safe, high-quality, and sustainable food and agricultural production worldwide.

3.12. Future prospects and recommendations

Looking ahead, the trajectory of HACCP, quality, and food safety management in food and agricultural systems is poised for transformative evolution, driven by technological advancements and increasing demand for minimally processed safe foods. AI and blockchain will revolutionize traceability, enabling predictive hazard detection and real-time supply chain oversight, potentially reducing recalls by 25 to 30% (Abobassier et al., 2026). Future outlook found in this study shows that by 2050, AI could reduce foodborne incidents by 20 to 30%, but human expertise remains central for integration, interpretation and ethical governance (McNamara, 2026). Also, more studies, e.g. primary research and systematic/scoping reviews, can be designed to synthesize and critically analyze tables under headings such as study quality, risk of bias, sample size, level of evidence, etc., to provide an additional layer of knowledge in HACCP and FSMS.

Substantial global disparities remain in the implementation and effectiveness of HACCP and integrated food safety management systems. High-income countries and large multinational enterprises have successfully adopted advanced digital tools, real-time monitoring, and risk-based frameworks aligned with ISO 22000:2018, resulting in superior hazard control and traceability. In contrast, low- and middle-income regions, along with small and medium enterprises (SMEs), continue to rely on traditional paper-based HACCP systems hampered by limited resources, inadequate infrastructure, and weak prerequisite programs (Insfran-Rivarola et al., 2020; USDA, 2025; McNamara, 2026).

Comparative analysis across standards reveals notable differences in rigor and applicability. BRCGS and IFS demand more stringent auditing and validation than Codex Alimentarius guidelines, while ISO 22000:2018 provides greater flexibility through operational prerequisite programs (Focusera Academy, 2024; Ali et al., 2026; Johnson et al., 2025). Religious and cultural certifications such as Halal further complicate harmonization efforts in multicultural supply chains (Fragkos et al., 2026; Varzakas & Antoniadou, 2024). Technology adoption also shows clear divides. While AI, IoT, and blockchain deliver strong results in developed contexts, their limited accessibility in resource-constrained settings widens the performance gap between global north and south actors (Insfran-Rivarola et al., 2020; USDA, 2025; McNamara, 2026). These findings indicate the need for more inclusive, scalable, and context-sensitive approaches to strengthen food safety management worldwide.

In environmentally vulnerable regions, adaptive HACCP models incorporating climate-resilient practices, such as drought-tolerant crop monitoring, will mitigate emerging risks, e.g. mycotoxin proliferation (Chavan et al., 2024). Efforts will continue to be made by researchers and industries toward improving HACCP and FSMS to mitigate the projected impacts of unsafe food, including AMR. Regulatory bodies such as the WHO advocate for One Health approaches, integrating human, animal, and environmental health to combat AMR (WHO, 2022). Future frameworks must harmonize standards across borders, with Codex Alimentarius updating guidelines for digital verification by 2030 (Codex Alimentarius Commission, 2020). Additional recommendations and prospects include directing improved training programmes for handlers in LMICs to bridge KAP gaps (knowledge, attitude, and practice gaps), targeting 50% increase in compliance (Insfran-Rivarola et al., 2020). Incentivizing sustainable innovations through subsidies to embed ERPs in HACCP will be helpful (USDA, 2025). Efforts are required for conducting longitudinal studies on ethical AI deployment to address biases in risk assessment. These steps will fortify resilient, inclusive systems, safeguarding public health amid uncertainties.

4. Conclusion

This systematic review synthesizes evidence from 2018 to 2026 on the evolution of HACCP, quality, and food safety management frameworks within increasingly complex food and agricultural systems. The findings demonstrate that while the foundational principles of HACCP remain robust, contemporary challenges, ranging from climate variability and antimicrobial resistance to globalized supply chains, demand more adaptive, integrated, and intelligent approaches. Significant progress is evident. The 2018 ISO 22000 revision has strengthened risk-based thinking and operational prerequisite programs, while digital technologies, AI, IoT, and blockchain are shifting FSMS from static, paper-based systems toward dynamic, predictive models. Integration of natural phytochemicals, Green HACCP principles, and enhanced on-farm risk management reflects a broader recognition that food safety must align with sustainability, quality, and resilience objectives. These advancements have delivered measurable improvements in hazard control, traceability, and operational efficiency, particularly in large-scale industrial settings. Nevertheless, persistent gaps remain. Small and medium enterprises, especially in developing regions, continue to face resource, technical, and knowledge barriers that limit effective implementation. Traditional HACCP application at the farm level remains fragmented, and the full potential of emerging technologies is yet to be equitably realized across diverse agri-food systems. Climate change further complicates hazard profiles, underlining the need for sensitive, flexible frameworks. Looking forward, improving HACCP and FSMS requires hybrid models that preserve preventive rigor while embracing data-driven innovation, human factors, and environmental stewardship. Policymakers, standard-setting bodies, and industry must prioritize capacity building, harmonized yet adaptable regulations, and inclusive technological access. Future research should focus on longitudinal impact evaluations, cost-effective solutions for SMEs, and the development of validated metrics for integrated safety sustainability performance. The next generation of food safety management must evolve beyond compliance toward a proactive, systems-oriented culture that safeguards public health, supports sustainable development, and builds resilience in an uncertain global food system.

This systematic review demonstrates that HACCP and broader FSMS have undergone substantial evolution since 2018, transitioning from static, compliance-oriented models toward dynamic, intelligent, and sustainability-integrated frameworks. Key advancements identified include the integration of digital technologies (IoT sensors, blockchain traceability, and AI-driven predictive analytics) for real-time hazard monitoring; the adoption of Green HACCP principles that embed environmental performance into core decision trees; the strategic use of phytochemicals and natural antimicrobials as sustainable biopreservatives; and enhanced, risk-based approaches to allergen and pathogen management supported by whole-genome sequencing and advanced detection tools.

These developments collectively strengthen proactive hazard control, improve supply chain transparency, and align food safety objectives with ecological imperatives. For industry practitioners, particularly large processors, the findings advocate investment in hybrid technological solutions and staff capacity building to maximize benefits while addressing implementation costs. SMEs, however, require tailored, scalable models and support mechanisms to avoid being left behind.

Policymakers should prioritize regulatory harmonization, updated Codex guidelines, and incentive structures that promote equitable digital and green transitions. Future research must focus on rigorous validation protocols for emerging controls, longitudinal impact assessments, and context-specific plans for developing economies. Realizing the full potential of these advancements will demand continued interdisciplinary collaboration to build resilient, safe, and truly sustainable global food systems.

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Data availability statement

Data sharing is not applicable to this article as all the data are sufficiently present here, and no additional data were created or analyzed in this study.

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